Draft version November 11, 2026

Typeset using LaTeX **twocolumn** style in AASTeX7.0.1



When Streams Curve Away: A Test of Dark Matter from Extragalactic Stellar Stream Populations

Nathaniel Starkman [1,2] Jacob Nibauer [3,4] Sarah Pearson [5,6] Sirui Wu [6] and Lina Necib [7,1]

[1] *Kavli Institute for Astrophysics and Space Research, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*
[2] *Case Western Reserve University, 10900 Euclid Ave, Cleveland, OH 44106, USA*
[3] *Department of Astrophysical Sciences, Princeton University, Princeton, NJ 08544, USA*
[4] *Center for Astrophysics | Harvard & Smithsonian, 60 Garden Street, Cambridge, MA 02138, USA*
[5] *DTU Space, Technical University of Denmark, Elektrovej 327, DK-2800 Kgs. Lyngby, Denmark*
[6] *Niels Bohr Institute, DARK, University of Copenhagen, Jagtvej 155A, 2200 Copenhagen N, Denmark*
[7] *Department of Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

ABSTRACT

The nature of dark matter sets the shape of galactic halos: collisionless cold dark matter (CDM) predicts triaxial halos, while scattering in self-interacting dark matter (SIDM) rounds them toward sphericity. We show that a large catalog of projected stellar stream tracks can statistically constrain halo shapes, and therefore the particle nature of dark matter, without detailed modeling of any individual stream or gravitational potential. We develop a new geometric diagnostic for stellar streams: if the curvature of any segment along a stream points away from the host galaxy's projected center of mass, then that stream falsifies broad classes of potentials, including spherical ones. We call such a segment convex. The diagnostic needs only a projected stream track and a host-center estimate, requiring no kinematics, distances, progenitor model, or fit to individual host potentials. Halo triaxiality sets the rate at which streams show convex segments, making this rate an observable of the halo-shape distribution and the underlying dark-matter model. We forecast that upcoming survey-scale stream morphology catalogs can discriminate between CDM and SIDM at up to 5σ.

Keywords: Stellar streams (2166) — Dark matter (353) — Cold dark matter (265) — Galaxy dark matter halos (1880) — Dark matter distribution (356) — Galaxy dynamics (591)

1. INTRODUCTION

In the cold dark matter (CDM) paradigm, halos are triaxial, with a shape distribution that depends on mass and redshift and with principal axes often misaligned with the baryons, e.g. the stellar disk where one is present (J. Bailin & M. Steinmetz 2005; C. A. Vera-Ciro et al. 2011; J. Baptista et al. 2023). In simulations, galactic halos have minor-to-major axis ratios $c/a \sim 0.5$–0.7, with weak dependence on environment (Y. P. Jing & Y. Suto 2002; B. Allgood et al. 2006; C. Giocoli et al. 2026). Baryons sphericalize halos more in the inner region, within $\sim 0.1\,R_{200} \approx 20$–$30\,\mathrm{kpc}$ for Milky-Way-mass systems (K. T. E. Chua et al. 2019; J. Prada et al. 2019; K. T. E. Chua et al. 2022; Q. Petit et al. 2023; D. Vargya et al. 2022). Here R_{200} is the radius enclosing a mean density 200 times critical density of the universe. Outside the inner regions halos remain more triaxial, with $c/a \sim 0.6$–0.7 and increasing triaxiality at $r \gtrsim 50\,\mathrm{kpc}$, retaining information about the initial cosmological collapse (S. Kazantzidis et al. 2004, 2010).

Alternative dark matter models predict different shapes. In this work we use Self-interacting dark matter (SIDM) as an example, however the methodology generalizes to other models. SIDM particles scatter and thermalize the inner halo, making the halo rounder than it would be in collisionless CDM (R. Davé et al. 2001; A. H. G. Peter et al. 2013; M. Kaplinghat et al. 2014). Halo shapes therefore probe the dark sector (for reviews see S. Tulin & H.-B. Yu 2018; S. Adhikari et al. 2025). Cosmological simulations with a dark matter self-interaction cross-section per unit mass of $\sigma/m \sim 1\,\mathrm{cm^2\,g^{-1}}$ predict measurable differences between SIDM and CDM axis ra-

Email: starkman@mit.edu

tios over much of the halo, from galaxies to clusters (D. Vargya et al. 2022; M. S. Fischer et al. 2024; T. Brinckmann et al. 2018). Baryons shrink this difference, so the shape signal is cleanest outside their dominant region (G. Despali et al. 2022).

However, measuring those shapes is hard. Weak gravitational lensing provides one way, through the distortion of background galaxies (M. Bartelmann & P. Schneider 2001). For $\sim 10^{11}$–$10^{13}\,M_\odot$ galaxies, however, the signal from an individual galaxy is too weak to provide a useful shape constraint, so weak-lensing analyses stack many systems, aligned by their light, to recover an ensemble-averaged shape (E. van Uitert et al. 2012; T. Schrabback et al. 2021). Stacking baryon-aligned systems can wash out signal as the outer halo orientation is not expected to couple to the baryons (P. Bett 2012); instead we want the ensemble shape information derived system by system.

Stellar streams, the tidal debris of disrupted globular clusters and dwarf galaxies, offer that information. They are excellent tracers of the gravitational potential because their stars occupy orbits near to their progenitors' (K. V. Johnston 1998; A. H. W. Küpper et al. 2012). Streams with large apocenters probe the outskirts of their host halos, where dark matter dominates the potential. These properties make stellar streams a promising test of SIDM halo shape, and SIDM zoom-in suites now provide galactic halo populations to predict halo shapes from (e.g. E. O. Nadler et al. 2025). In the Milky Way, 6D astrometry has enabled detailed stream-based constraints on the Galactic potential and its dark matter halo (D. R. Law & S. R. Majewski 2010; S. E. Koposov et al. 2010; J. Bovy et al. 2016; A. Bonaca & D. W. Hogg 2018; E. Vasiliev et al. 2021; R. Ibata et al. 2024; A. Bariego-Quintana & F. J. Llanes-Estrada 2024).

Beyond the Local Group, surveys cannot recover full 6D phase-space information for essentially any stream (D. Martínez-Delgado et al. 2010), and cannot measure the three-dimensional stream torsion that encodes halo asphericity (A. Bariego-Quintana & F. J. Llanes-Estrada 2024). Current and upcoming wide-field imaging surveys will not measure these quantities either. The *Euclid* mission (Euclid: Aussel, H. et al. 2025; Euclid: Walmsley, M. et al. 2025), the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST) (Ž. Ivezić et al. 2019), *ARRAKIHS*[8] (R. Guzman et al. 2022), and the *Nancy Grace Roman Space Telescope* (D. Spergel et al. 2015) will instead provide large stream catalogs. Any one projected track typically constrains its host's shape weakly, so the statistical power comes from measuring many systems. We therefore need a homogeneous method that works on projected tracks, constrains each system separately, and combines these individual constraints at the population level without assuming that each halo is aligned with its galaxy's baryons. Stream catalogs spanning a wide range of host masses, morphologies, and redshifts are now arriving from Dark Energy Survey (DES), the Stellar Streams Legacy Survey, and other facilities (J. Miró-Carretero et al. 2024, 2025; D. Martínez-Delgado et al. 2025; E. Sola et al. 2025). Automated pipelines can scale detection to the full survey footprint (M. Walmsley et al. 2023; A. J. Gordon et al. 2024). With track fits alone, Euclid: Starkman, N. et al. (2026) individually and jointly constrained host-halo shapes from 13 streams in *Euclid* Q1 imaging.

Existing methods trade fidelity against assumptions, from full N-body models down to orbit fits. Full N-body forward modeling captures tidal disruption in detail but requires expensive computation and strong priors on the progenitor and orbital history (M. A. Fardal et al. 2013); particle-spray models reduce that cost at the price of more prescriptive progenitor and tidal-release assumptions (M. A. Fardal et al. 2015; S. Pearson et al. 2022), and with GPU acceleration can search over halo parameters for external galaxies (J. Nibauer & S. Pearson 2026; D. Chemaly et al. 2026a). Orbit-fitting approaches are cheaper still (M. Walder et al. 2025), but identify the stream with an orbit, a known source of bias (J. L. Sanders & J. Binney 2013a) that angle-action modeling of the full stream distribution removes (J. L. Sanders & J. Binney 2013b). The curvature-based method of J. Nibauer et al. (2023), hereafter N23, requires far fewer assumptions, inferring the potential's shape from projected curvature, since a stream's curvature points within 90° of the projected acceleration. This method is applicable to large samples of stellar streams. S. Wu et al. (2026) applied it to nearby galaxies, and Euclid: Starkman, N. et al. (2026) to deeper *Euclid* imaging. Both infer halo shapes ranging from spherical to CDM-consistent flattening, using samples of tens of streams, as does the population-level particle-spray modeling of D. Chemaly et al. (2026b).

Stream-based methods that assume a specific host potential and a prescription for the progenitor's structure and history are informative on a per-system basis, but at great risk of systematic errors. Large stream catalogs make the opposite end of that trade viable, where a diag-

[8] Analysis of Resolved Remnants of Accreted galaxies as a Key Instrument for Halo Surveys.

nostic assumes neither a specific potential nor a stream model and is informative only in aggregate. We develop that test of galactic gravitational potentials based on the purely geometric diagnostic of the detection, or non-detection, of stellar stream segments that exhibit curvature away from the galaxy's projected center of mass (CoM). Geometrically, convexity of the curvature (truly non-concavity: a straight track, or a convex one curving away from the CoM) is the projection of the orbit's torsion, where an aspherical potential twists the intrinsic track out of the plane containing the CoM. This extends the torsion diagnostic of A. Bariego-Quintana & F. J. Llanes-Estrada (2024) from 3D in the Galaxy to 2D in extragalactic systems. A track concave toward the CoM is consistent with axisymmetry about the line-of-sight (LoS); a convex track rules it out.

Because convexity provides a one-sided constraint for an individual stream, we use the diagnostic statistically: we predict the convexity rate in CDM and SIDM halo populations and determine how many streams are needed to distinguish between the predictions. The fraction of a halo's streams with convexities decreases as the halo becomes rounder, and is zero for a spherical halo. This fraction therefore traces host-halo shape, and CDM and SIDM predict different halo-shape distributions. The convexity rate reflects both halo shape and the model-dependent orbital histories of stream progenitors (E. O. Nadler et al. 2020).

Each stream contributes a yes-or-no observation of whether it contains a convex segment. Together, these observations make the catalog a survival test rather than a fit: a dark-matter model must account for the observed number of streams with convex segments. The diagnostic needs only the projected track and the projected CoM, so it suits large and heterogeneously-processed subsamples from *Euclid*, Rubin, and related surveys.

We organize the paper as follows. Section 2 motivates the diagnostic with a mock observation of GD-1 and builds the curvature-alignment statistic from the projected on-sky geometry. Section 3 derives when orbits can be convex and carries the criterion to streams. Section 4 applies it to constrain individual potentials, confirming the halo triaxiality bounds numerically against particle-spray and N-body streams. Section 5 builds population-level hypothesis tests and derives the sample sizes needed to distinguish CDM from SIDM at survey scale. Section 6 covers observational implications and Section 7 summarizes.

2. OBSERVING STREAMS AROUND GALAXIES

We begin with a mock observation of GD-1 (C. J. Grillmair & O. Dionatos 2006), a stellar stream whose well-measured track we can reproject as if viewed from outside the Galaxy (K. Malhan et al. 2018; N. Starkman et al. 2025). At some viewing angles the projected GD-1 stream appears convex. We use this behavior to motivate the diagnostic we develop in the following sections.

2.1. *Mock-Observing GD-1*

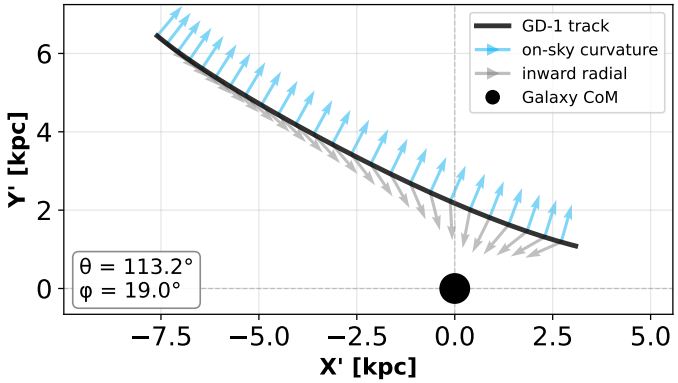


Figure 1. Mock observation of the GD-1 stream: the stream's ridge line (black) reprojected onto the sky plane as if viewed from outside the Galaxy, at the viewing angle (θ, ϕ) (lower left) that maximizes the convex fraction of the track. Cyan arrows show the on-sky curvature direction; gray arrows show the inward radial direction, toward the projected CoM (black dot at the origin). The curvature points away from the CoM along the entire track; the track is everywhere convex. This convexity is inconsistent with any static, centrally-concentrated potential axisymmetric about the viewing direction, including all spherical potentials.

Before developing any formalism, we show that stream convexities exist in our own Galaxy. We place an observer at infinity, take the A. M. Price-Whelan & A. Bonaca (2018) track from the `galstreams` package (C. Mateu 2023), and project it onto the sky plane.

We project GD-1 along 10^4 isotropically distributed lines of sight. Of these projections, 23.1% contain a convex segment, 3.2% contain one longer than 10 kpc, and 1.6% are convex along the full track. For illustration, Figure 1 shows the view from Galactocentric LoS $(\theta, \phi) = (113.2°, 19°)$, where θ is measured from the north Galactic pole and ϕ from the Sun-to-center direction. In this projection, GD-1 is well separated from the projected CoM, and its curvature points away from the Galactic center along 100% of its arc length. The cyan arrows show this projected curvature, while the gray arrows show the inward radial direction. The two point in

opposite directions even though the track curves toward the Galactic center in 3D. This convexity is plain by eye. As we show below, a convex segment rules out any static, centrally concentrated potential that is axisymmetric about the LoS, including a spherical potential. This test is a projected counterpart to the torsion diagnostic of A. Bariego-Quintana & F. J. Llanes-Estrada (2024), which uses three-dimensional stream tracks to detect asphericity. The result for GD-1 is unsurprising because the Milky Way is neither spherical nor static: it contains a disk and bar (J. Bland-Hawthorn & O. Gerhard 2016), has a triaxial dark matter halo (D. R. Law & S. R. Majewski 2010; C. Vera-Ciro & A. Helmi 2013; J. J. Han et al. 2022), and moves in response to the Large Magellanic Cloud (N. Garavito-Camargo et al. 2021). The example nevertheless demonstrates that convex stream segments occur and can be identified from projected tracks.

2.2. *The Observer*

To turn an observed convexity into a constraint on the host potential, we work in terms of what an observer can measure. Wide-field imaging surveys such as *Euclid*, Rubin/LSST, and *Roman* measure detailed on-sky positions; host redshifts are often available, but proper motions are not. Stream kinematics have been measured for only a handful of systems: the M31 Giant Stream (R. Ibata et al. 2004), the NGC 4449 stream (E. Toloba et al. 2016a), and a disrupting dwarf in NGC 253 (E. Toloba et al. 2016b). Methods combining deep imaging with planetary-nebula kinematics may extend such measurements (L. M. Valenzuela et al. 2026), but on-sky positions remain the primary observable for large samples. We therefore work in an observer's frame (Figure 2), measuring primarily the on-sky image.

To describe the galaxy we define a coordinate system $\boldsymbol{X} = (X, Y, Z)$ intrinsic to the inner galaxy, with origin at the CoM of the stellar distribution and orientation aligned to the galaxy (e.g. $\hat{Z}$ parallel to a disk's normal). The relevant components are the inner galaxy (e.g., a baryonic disk), the dark matter halo, and any massive satellite systems. The observer lies on the $+\hat{\boldsymbol{e}}_z$ axis of a *viewing* frame (x, y, z) (sky plane (x, y)), related to the intrinsic frame by the general orthogonal transformation $(x, y, z) = \boldsymbol{Q}\ (X, Y, Z)$, e.g. a rotation matrix.

For any viewing-frame vector $\boldsymbol{u} = (u_x, u_y, u_z)$ we define the sky-plane projection $\mathrm{pr}(\boldsymbol{u}) \equiv (u_x, u_y)$. We also use $(\cdot)_p$ for sky-plane quantities, which are not always simply projections from 3D. The sky-plane CoM is $\bar{\boldsymbol{x}}_p \equiv \mathrm{pr}(\bar{\boldsymbol{x}}) = {M_{\rm tot}}^{-1} \int_V \mathrm{pr}(\boldsymbol{x})\ \rho(\boldsymbol{x}, t)\, \mathrm{d}^3\boldsymbol{x}$, i.e. the mass-weighted mean of $\boldsymbol{x}_p = (x, y)$, marked in Figure 2.

2.3. *The Observed Curve*

Figure 2 shows the geometry of an observed track curve. We develop it for a general curve here and apply it to an orbit in Section 3 and to a stream ridge line in Section 3.2. We parametrize the path by the planar arc length s_p: $\boldsymbol{x}_p(s_p) = (x(s_p), y(s_p))$. The unit tangent to the planar path is

$$\hat{\boldsymbol{t}}_p(s_p) = \frac{\mathrm{d}\boldsymbol{x}_p}{\mathrm{d}s_p}, \qquad \|\hat{\boldsymbol{t}}_p\| = 1. \tag{1}$$

The planar curvature $\kappa_p(s_p)$ measures how rapidly the planar tangent direction changes along the planar path:

$$\frac{\mathrm{d}\hat{\boldsymbol{t}}_p}{\mathrm{d}s_p} = \kappa_p(s_p)\, \hat{\boldsymbol{n}}_p(s_p), \qquad \kappa_p(s_p) = \left\| \frac{\mathrm{d}\hat{\boldsymbol{t}}_p}{\mathrm{d}s_p} \right\|, \tag{2}$$

$$\|\hat{\boldsymbol{n}}_p\| = 1, \quad \hat{\boldsymbol{t}}_p \cdot \hat{\boldsymbol{n}}_p = 0. \tag{3}$$

where $\hat{\boldsymbol{n}}_p$ is the unit normal to the planar path in the sky plane.

To relate the curvature normal direction to the projected CoM, we define a dimensionless curvature alignment diagnostic for any planar curve.

$$\mathcal{A} \equiv -\mathrm{sign}\left(\hat{\boldsymbol{n}}_{\rm p} \cdot \boldsymbol{x}_{\rm p}\right). \tag{4}$$

The colored regions in Figure 2 show this: $\mathcal{A} = +1$ means the curvature normal points "inward" toward the stellar CoM $\bar{\boldsymbol{x}}_p$, labeling the curve as concave (green region). Conversely, $\mathcal{A} = -1$ means it points "outward," labeling the curve as convex (red region). The diagnostic is undefined when $\boldsymbol{x}_p = \boldsymbol{0}$ (the galaxy's CoM), when the curve is entirely radial, or when $\kappa_p = 0$ (an inflection point or locally straight segment). In the last case we extend the classification by continuity. A straight segment does not curve toward the CoM, so we assign it to the convex class ($\mathcal{A} \leq 0$). A straight segment that is itself radial falls under both cases, where the radial classification takes precedence and $\mathcal{A}$ remains undefined (Section 3).

3. THE STREAM CONVEXITY DIAGNOSTIC

In this section we develop the convexity diagnostic as a test of the potential. We first determine, directly from the projected equations of motion, when a potential can host orbits with convex-in-projection segments ($\mathcal{A} \leq 0$). We then specialize it to LoS-axisymmetric centrally-concentrated potentials, and carry the orbit-level criterion onto the observable stream ridge line.

We work in the observer-aligned viewing frame centered on the galaxy's CoM, illustrated in Figure 2 and described in Section 2.2. We identify this stellar CoM

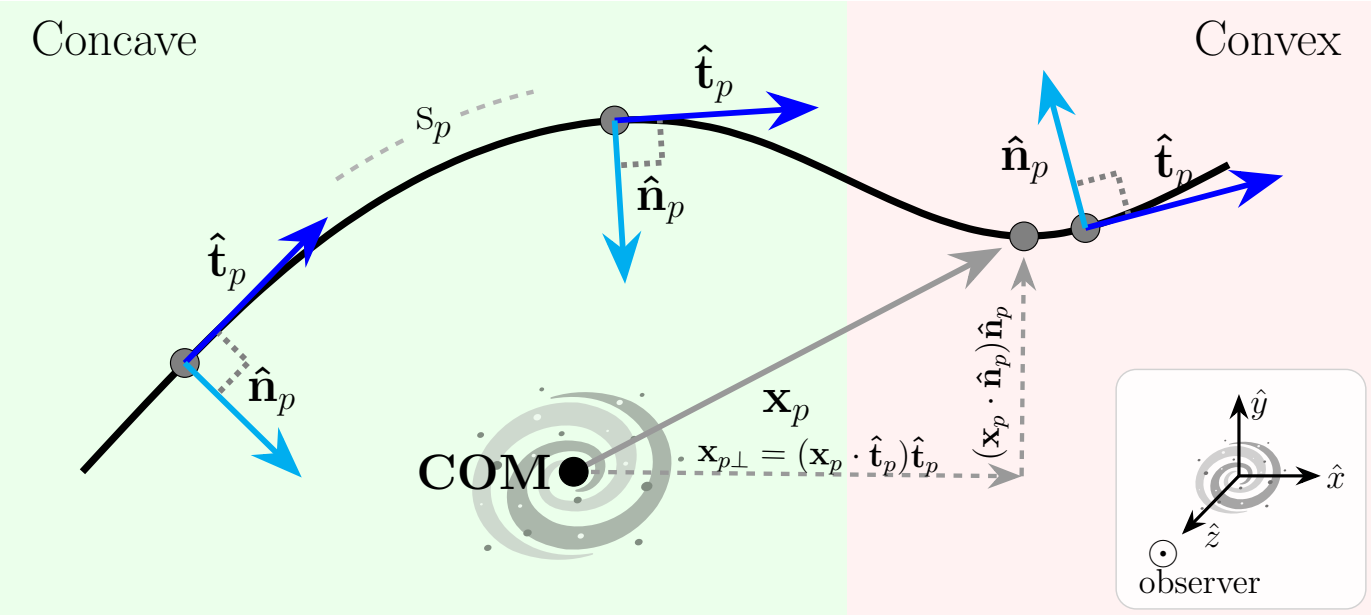


Figure 2. Geometry of the on-sky curvature alignment diagnostic, shown in observer (sky-plane) coordinates. The black curve is an observed projected path $\boldsymbol{x}_p(s_p)$ of a stellar stream, parametrized by projected arc length s_p (gray dashed indicator). A marker shows the location of the galaxy's projected stellar CoM. Blue and cyan arrows mark the unit tangent $\hat{\boldsymbol{t}}_p$ and unit normal $\hat{\boldsymbol{n}}_p$, respectively, at points along the curve. The gray arrow $\boldsymbol{x}_p$ runs from the CoM to a point on the curve and decomposes into its tangential $(\boldsymbol{x}_p \cdot \hat{\boldsymbol{t}}_p)\hat{\boldsymbol{t}}_p$ and normal $(\boldsymbol{x}_p \cdot \hat{\boldsymbol{n}}_p)\hat{\boldsymbol{n}}_p$ components (dashed). The sign of $\boldsymbol{x}_p \cdot \hat{\boldsymbol{n}}_p$ defines the alignment $\mathcal{A} \equiv -\mathrm{sign}(\hat{\boldsymbol{n}}_\mathrm{p} \cdot \boldsymbol{x}_\mathrm{p})$ (equation (4)): where $\hat{\boldsymbol{n}}_p$ points toward the CoM, the curve is concave ($\mathcal{A} = +1$, green); where it points away, the curve is convex ($\mathcal{A} = -1$, red).

with the center of the potential, the origin for the radial force $F_R = \partial_R \Phi$. The dark matter center is expected to coincide closely with the stellar CoM: >95 % of *Eagle* galaxies have offsets below 700 pc (M. Schaller et al. 2015), Milky-Way analogs have offsets of a few hundred parsec (M. Kuhlen et al. 2013), and cluster brightest-galaxies have median offsets of ∼1 kpc (C. Roche et al. 2024). Assuming the stellar CoM is the potential center is the one assumption we make about the potential and is weaker than adopting a halo profile or a shape. Also, while we defer this to a future work, jointly exploring the potential center with the data is a straightforward extension to this paper (Section 6.2). In this view frame, in cylindrical coordinates (R, θ, z), with z along the LoS, the projected orbit is the planar curve $(R(\theta), \theta)$.

For an orbit in a potential $\Phi(R, \theta, z)$ the projected equations of motion are

$$\ddot{R} - R\dot{\theta}^2 = -\partial_R \Phi, \qquad 2\dot{R}\dot{\theta} + R\ddot{\theta} = -\frac{1}{R}\partial_\theta \Phi, \tag{5}$$

where we will interchangeably notate ∂_x and $\frac{\partial}{\partial x}$. The projected orbit is the plane curve $R(\theta)$ traced by $(R(t), \theta(t))$. Equation (5) is exact since the cylindrical decomposition separates cleanly in z.

From equation (2), the curvature of a polar curve $R(\theta)$ is

$$\kappa_p = \left| \frac{R^2 + 2R'^2 - RR''}{\left(R^2 + R'^2\right)^{3/2}} \right|, \tag{6}$$

where

$$R' \equiv \partial_\theta R, \; R'' \equiv \partial_\theta^2 R. \tag{7}$$

Since equation (6) is a geometric property of the curve $R(\theta)$, it requires no assumption about the direction of the orbit.

Differentiating $R' = \dot{R}/\dot{\theta}$ gives

$$\dot{\theta}^3 R'' = \ddot{R}\dot{\theta} - \dot{R}\ddot{\theta}. \tag{8}$$

Substituting equation (8) and the equations of motion (equation (5)) into the numerator of equation (6), leaves

$$R^2 + 2R'^2 - RR'' = \frac{1}{\dot{\theta}^2}\left(R F_R - R' F_\theta\right), \tag{9}$$

where $F_\theta = R^{-1}\partial_\theta \Phi$. Since $\dot{\theta}^2 > 0$, the projected orbit is convex if and only if locally

$$\boxed{R F_R \leq \frac{\partial R}{\partial \theta} F_\theta,} \tag{10}$$

with strict inequality giving convexity and equality giving a straight segment. The left side of equation (10),

$R\,F_R$, is the radially-weighted radial force. For mass profiles including a Keplerian potential, it pulls the track concavely toward the CoM. The right side is the azimuthal force, weighted by the local projected slope R', partly set by how plunging the orbit is. Where the projected track is nearly radial, e.g. for an eccentric, radially plunging orbit, the azimuthal term is amplified, so even a weak azimuthal perturbation can produce a convexity.

Projected convexity can be sourced from 3D torsion. In an aspherical potential orbits experience nonzero torsion, which twists their osculating planes out of the plane containing the CoM (A. Bariego-Quintana & F. J. Llanes-Estrada 2024). Projected onto the sky, this torsion can reverse the sign of the apparent curvature. Torsion gives the 3D picture, but we work from the projected equations of motion to derive the convexity threshold explicitly.

Equivalently, equation (10) can be interpreted as a condition on the projected acceleration along the path. That acceleration is the gravitational field $\boldsymbol{a}_p = -F_R\,\hat{R} - F_\theta\,\hat{\theta}$. Resolving $\boldsymbol{a}_p$ relative to the projected velocity $\boldsymbol{v}_p$ (speed $v_p \equiv \|\boldsymbol{v}_p\|$), only its perpendicular part $\boldsymbol{a}_{p\perp} \equiv \boldsymbol{a}_p - (\boldsymbol{a}_p \cdot \hat{\boldsymbol{t}}_p)\hat{\boldsymbol{t}}_p$ bends the track. This perpendicular acceleration fixes the curvature normal through $\kappa_p\,\hat{\boldsymbol{n}}_p = \boldsymbol{a}_{p\perp}/v_p^2$. Substituting into equation (10), the kinematic prefactors cancel and the convexity condition reduces to the sign of the projected acceleration toward the CoM,

$$\mathcal{A} = -\mathrm{sign}\left(\boldsymbol{a}_{p\perp} \cdot \boldsymbol{x}_p\right). \tag{11}$$

The criterion depends on $\boldsymbol{a}_p$ at that location, not on the speed v_p. This speed-independence underlies the curvature method of N23: the shape of the projected curve alone constrains the projected acceleration field. With the direction-independence of equation (6), we need only on-sky positions.

3.1. *Ruling Out LoS-Axisymmetric Centrally-Concentrated Mass Profiles*

For a potential axisymmetric about the LoS, $\partial_\theta \Phi = 0$ in observer-aligned frames, so the azimuthal-force term on the right-hand side of equation (10) vanishes. The convexity condition then reduces to $F_R \leq 0$, a net outward radial force. Centrally-concentrated potentials, such as the Navarro–Frenk–White (NFW) profile (J. F. Navarro et al. 1996, 1997), have $\partial_R \Phi > 0$ everywhere except at the CoM, so nothing satisfies this condition. Therefore, in any LoS-axisymmetric centrally-concentrated potential, the projected orbit is always concave toward the CoM ($\mathcal{A} = +1$), independent of inclination and eccentricity. Spherical potentials $\Phi(r)$ are the trivial case: every axis through the CoM is a symmetry axis, so this holds for any viewing geometry. A convex segment therefore requires an azimuthal force in the observer's frame, either intrinsic (e.g., a triaxial halo or a bar) or induced by off-axis viewing an axisymmetric potential.

The diagnostic fails in only one case, a tangent-passage, where the projected tangent points straight at the projected CoM ($\boldsymbol{x}_{p\perp} = \boldsymbol{0}$) and $\mathcal{A}$ vanishes. The angular momentum about the LoS, $L_{\rm los} = R^2\dot{\theta}$, sets how close the tangent line comes to the CoM: the perpendicular distance is $\|\boldsymbol{x}_{p\perp}\| = R^2|\dot{\theta}|/v_p = |L_{\rm los}|/v_p$. A tangent-passage therefore means $L_{\rm los} = 0$. In a LoS-axisymmetric potential $L_{\rm los}$ is conserved, so this is all-or-nothing: either the orbit projects to a radial line through the CoM and $\mathcal{A}$ is undefined along its whole length, or the tangent line never reaches the CoM and $\mathcal{A}$ is defined everywhere. The first case looks like a straight track pointing at the CoM, and we exclude such tracks from the diagnostic. It is the only straight projected track allowed by LoS-axisymmetry, and so the only exception to the straight-segment convention of Section 2.3.

Away from this geometry, observation of a single convex arc ($\mathcal{A} \leq 0$) anywhere along an observed track rules out the whole class: any spherical potential, and more generally any LoS-axisymmetric centrally-concentrated one.

3.2. *Applying the Diagnostic to Streams*

The criterion so far applies to an orbit, but a stream is not an orbit. It is a collection of stars stripped at different times onto nearby orbits, and we observe the ridge line they trace (Figure 2). We evaluate the diagnostic on that stream ridge line, so we need to know when the ridge line and the orbits composing it have the same concavity.

Let $\hat{\boldsymbol{t}}_p^{\rm str}$ be the stream (ridge-line) tangent (equation (1)) and $\hat{\boldsymbol{t}}_p^{\rm orb}$ the tangent of the orbit of the star at that point. The angle between them is

$$\delta\theta(s_p) \equiv \arccos\left(\left|\hat{\boldsymbol{t}}_p^{\rm str}(s_p) \cdot \hat{\boldsymbol{t}}_p^{\rm orb}(s_p)\right|\right). \tag{12}$$

Equation (2) fixes the normal to the tangent, so the same angle separates the two curvature normals. Where $\delta\theta$ is small, the ridge line and the orbit have the same $\mathcal{A}$.

How fast the stream widens sets that angle $\delta\theta$. Let $W(s_p)$ be the root-mean-square (RMS) transverse

spread of stars about the ridge line, set by the release-velocity dispersion smearing the debris into a band (A. H. W. Küpper et al. 2010, 2012; J. Bovy 2014). Over a short arc length δs_p the ridge line is locally straight, but its edge moves by δW, so the edge crosses the ridge direction at an angle $\tan \delta\theta = \delta W / \delta s_p$. Where this angle is large the orbits point, and so curve, away from the ridge line. Therefore, even for relatively wide streams, it is where the width changes rapidly that the ridge line and orbits can have different concavities; elsewhere they agree.

The stream is within the transverse offsets $\pm W(s_p)$ from the ridge line, and has slopes relative to the ridge line bounded by $|\partial_{s_p} W|$. The tangent mismatch therefore satisfies

$$\delta\theta(s_p) \lesssim \left|\partial_{s_p} W(s_p)\right|. \tag{13}$$

Section A.1 derives similar bounds for every track within the stream, not only for its edges. The tangent mismatch therefore depends on the width gradient more than the width itself: even a wide stream has well-aligned tangents where its width is constant.

Whether $\delta\theta$ flips the sign of $\mathcal{A}$ also depends on $\|\boldsymbol{x}_{p\perp}\|$, the perpendicular distance from the projected CoM to the tangent line. Equation (4) defines the alignment; rotating the normal by $\delta\theta$ changes that product by $\|\boldsymbol{x}_p\| \sin\delta\theta$, so the sign holds only where

$$\|\boldsymbol{x}_{p\perp}\| \gg \|\boldsymbol{x}_p\| \sin\delta\theta. \tag{14}$$

The left-hand side is small where the track points near the CoM. This is the tangent-passage of Section 3, where even small $\delta\theta$ can reverse $\mathcal{A}$. We call the band around it in which equation (14) fails the *near-tangent strip*, and exclude it from the diagnostic. With equation (13), the ridge-line alignment matches that of its constituent orbits where

$$\boxed{\|\boldsymbol{x}_{p\perp}\| \gg \|\boldsymbol{x}_p\| \left|\partial_{s_p} W(s_p)\right|.} \tag{15}$$

Every quantity is measurable from imaging and does not require assumptions relating the width to the orbital scale. Where the width varies by $|\partial_{s_p} W| \sim W/\|\boldsymbol{x}_p\|$, the condition becomes $\|\boldsymbol{x}_{p\perp}\| \gg W$, i.e. the projected CoM must lie outside a strip of width comparable to the stream width from the local tangent line. These derivations are geometric and independent of how the debris was released, relying instead on continuity and coherence arguments intrinsic to non-disrupting streams.

Empirically, GD-1 and Pal 5 have RMS widths $W \sim$ 50–150 pc over tens of kpc (S. E. Koposov et al. 2010; D. Erkal et al. 2017; A. Bonaca et al. 2020), and dwarf-galaxy streams reach $W \sim$ 2–3 kpc (Sagittarius, M. Correnti et al. 2010; Centaurus A, S. Pearson et al. 2022) over comparable or greater lengths, giving $|\partial_{s_p} W| \sim W/L \lesssim 10^{-2}$. The excluded near-tangent strip, $|\boldsymbol{x}_{p\perp}| \lesssim W$, is therefore 10–100 times wider for dwarf-galaxy streams than for globular-cluster streams, so for the latter the diagnostic applies closer to edge-on viewing angles and to more radial projected tracks. Globular-cluster streams are the preferable system, but very few are known beyond the Local Group (J. K. Holm et al. 2026) and the broader dwarf-galaxy streams dominate extragalactic catalogs. The curvature criterion of N23 requires a similar agreement between tracks and orbit, there emphasizing coherence as a survival condition of the stream to bound the track-orbit angles. Here we further require the angular offsets not to change $\mathcal{A}$ between track and orbits. Differences of order the stream width therefore do not change the sign of $\mathcal{A}_{\rm str}$ (Section A.2). Where $|\partial_{s_p} W|$ is large $\mathcal{A}_{\rm str}$ can become unreliable: in fanning regions near apocenter or after a strong tidal interaction, and in the shells and caustics formed by low-angular-momentum debris (D. Hendel & K. V. Johnston 2015). For an example in the Milky Way, parts of Pal 5's thin track lie in a fanning region (N. Starkman et al. 2020; S. Pearson et al. 2015) and would be unusable. Also, within a few tidal radii of an intact progenitor $\Phi_{\rm prog}$ will shift the curvature normal. All three are identifiable in imaging, so we should evaluate $\mathcal{A}_{\rm str}$ only on segments well separated from the progenitor where $W(s_p)$ varies slowly. For the rest of this work we therefore apply the orbit criterion of equation (11) directly to observed stream tracks, with $|\partial_{s_p} W|$ as the check on where it applies. The ridge line inherits the orbit's curvature sign wherever this width bound holds, so the exclusion of Section 3.1 applies to the observed stream track outside the near-tangent strip.

4. CONVEXITIES IN TRIAXIAL POTENTIALS

Convex segments can occur in a triaxial potential from certain viewing directions, which we establish using numerical orbits and the general form of $\mathcal{A}$ in Section 4.1. Last, in Section 4.2 we confirm the results of Section 3.2 and show that convexities occur in simulated streams.

4.1. *Orbits in Triaxial Potentials*

No halo is perfectly axisymmetric, so ruling out that class (Section 3.1) is only a starting point. Here, we investigate how a triaxial potential produces convexities, deriving the alignment $\mathcal{A}$ for a general triaxial potential and showing convexities in numerical orbit solutions.

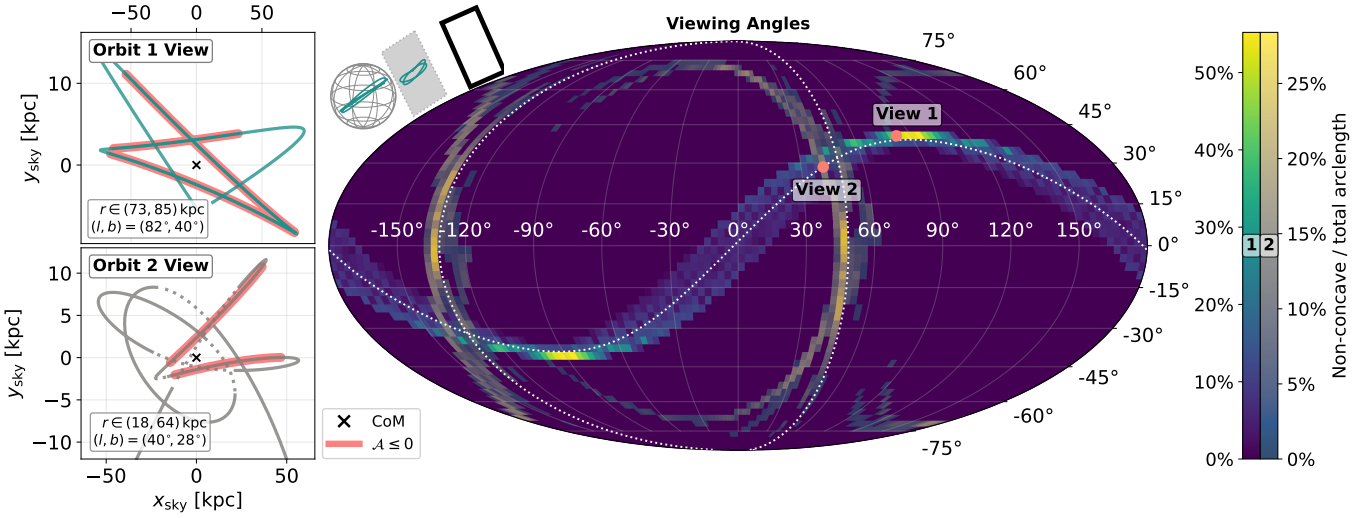


Figure 3. *Fraction of a projected orbit that appears convex, across viewing angles, for two orbits in a triaxial NFW potential. Left*: sky-plane projections of Orbit 1 (top, $e \in [0.06, 0.08]$) and Orbit 2 (bottom, $e \in [0.48, 0.57]$), each shown at the viewing angle specified with a red dot in the right plot. Red segments mark the convex ($\mathcal{A} \leq 0$) portions; the cross denotes the CoM. Dotted segments show where Orbit 2 (never Orbit 1) passes within 25 kpc of the CoM, where baryonic contributions are more significant. *Right*: Mollweide projections over all viewing directions for Orbit 1 (viridis, labeled 1) and Orbit 2 (cividis, labeled 2). At each pixel, the color encodes the fraction of the projected orbit that appears convex when viewed from that direction. White dotted great circles mark the mean orbital plane of each orbit; viewing directions lying on these circles see parts of the orbit edge-on. The brightest regions are the viewing angles at which the largest convex fraction is visible.

Figure 3 shows two test-particle orbits integrated for 5 Gyr in a pure triaxial density NFW potential ($M = 10^{12}\,\mathrm{M}_\odot$, $r_s = 14\,\mathrm{kpc}$, $b/a = 0.85$, $c/a = 0.7$; no disk component), at low and moderate eccentricity (Orbit 1, $e \in [0.06, 0.08]$; Orbit 2, $e \in [0.48, 0.57]$). The triaxial NFW density model (J. F. Navarro et al. 1996, 1997) is the most relevant triaxial density profile for CDM dark matter halos. We hold each orbit fixed while the viewing direction varies: for each direction we project the orbit onto the observer's sky and measure the projected arc-length fraction that is convex. The left panels show each orbit as it appears from the viewing direction marked in the right panel, with the convex arcs in red. The right panel maps this convex fraction over viewing direction in Mollweide projection, darkest where no segment is convex. Both orbits lie predominantly beyond 25 kpc, well into the dark-matter-dominated regime for galaxies. The rare interior segments (dotted) do not overlap the convex arcs — the convexities are not caused by an inner galaxy component like a disk, which this model does not include.

From most viewing directions the two orbits are entirely concave; at some viewing angles segments become convex. The fraction of each orbit that is convex at a given viewing angle reaches $\sim 55\%$ for Orbit 1 and $\sim 28\%$ for Orbit 2. Most orbit segments that become convex do so when viewed near edge-on to the orbit's mean orbital plane (dotted white lines). Face-on views, perpendicular to the orbital plane, show almost no orbit convexities. Geometrically, this is expected. Torsion from the triaxial potential pulls orbits out of a fixed orbital plane, and so convex deviations from great circles are largest here, where the torsion signals are most evident (A. Bariego-Quintana & F. J. Llanes-Estrada 2024). Within each orbit, the intersections of its great-circle bands mark viewing directions where multiple convex segments of that same orbit are simultaneously visible.

We now build some analytic understanding of orbit convexities in triaxial potentials. Figure 3 showed convexities in triaxial NFW density profiles; however from equation (10) we understand that convexities arise more straightforwardly from the potential. As further explained in Appendix B, triaxial densities do not generically have simple analytic parametrizations in the potential, but are nearly the same as a triaxial potential up to small correction terms for realistic halos. We therefore consider triaxial potentials.

With the convexity diagnostic we can build analytic understanding and constraints on triaxial potentials without ever assuming a specific potential parametrization like an NFW model. We only require the generic 3D form $\Phi(\boldsymbol{x}) = \Phi(m)$ with

$$m^2 \equiv \boldsymbol{x}^{\mathrm{T}} \boldsymbol{M} \boldsymbol{x}, \qquad \boldsymbol{M} = \begin{pmatrix} \boldsymbol{M}_{pp} & \boldsymbol{M}_{pz} \\ \boldsymbol{M}_{pz}^{\mathrm{T}} & M_{zz} \end{pmatrix}, \tag{16}$$

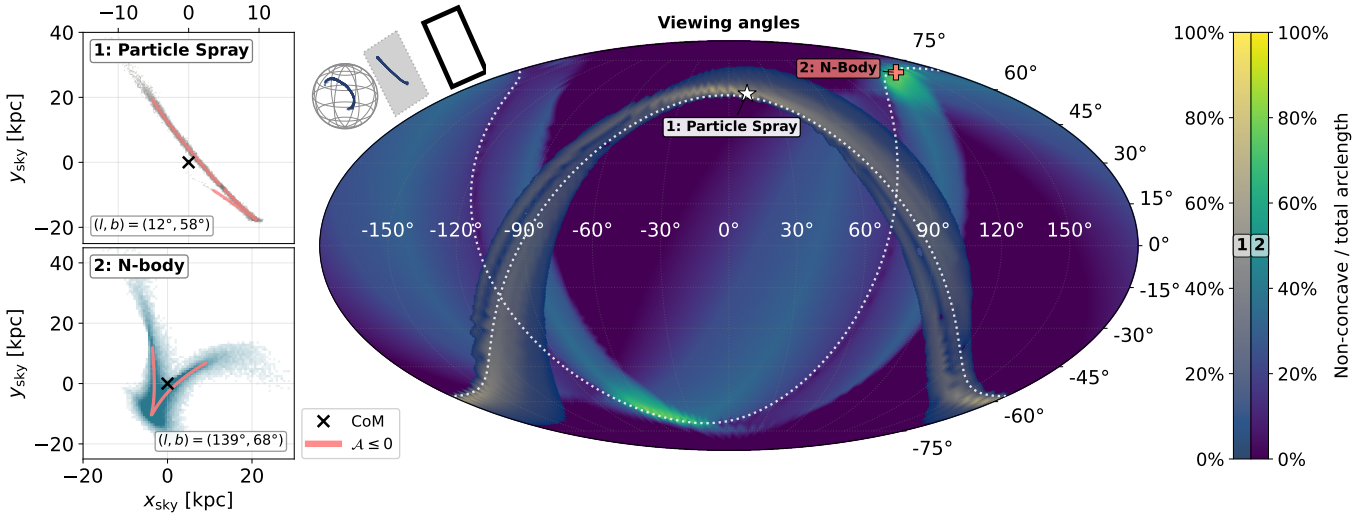


Figure 4. *Convex arc-length fraction of two different kinds of simulated streams, across viewing angles.* The particle spray simulation uses the same initial condition and triaxial NFW potential as Figure 3, while the N-body comes from N23. *Left*: sky-plane projections of particle-spray (top) and N-body (bottom), each shown at the viewing angle specified with a white star and an orange cross in the right plot. Red segments mark the convex portions; the black cross denotes the CoM. *Right*: Mollweide projections over all viewing directions for particle-spray simulation 1 (cividis, labeled 1) and N-body simulation 2 (viridis, labeled 2). Panels are otherwise as in Figure 3, which shows convexities in orbits. This figure extends that result to streams, finding visible convexities in particle-spray and N-body streams.

where the symmetric shape matrix $\boldsymbol{M}$ is written in observer-aligned coordinates, $\boldsymbol{x} = \boldsymbol{x}_p + z\hat{\boldsymbol{e}}_z$, with sky-plane block $\boldsymbol{M}_{pp}$, sky–LoS column $\boldsymbol{M}_{pz}$, and LoS entry M_{zz}. Applying $\boldsymbol{Q}^{-1}$ to transform into the principal-axis frame, this reduces to the much simpler $\boldsymbol{M} = \mathrm{diag}(a_\Phi^{-2}, b_\Phi^{-2}, c_\Phi^{-2})$, where $a_\Phi \geq b_\Phi \geq c_\Phi$ are the semi-axes of the isopotentials, which are rounder than the density (Appendix B). The acceleration field is

$$\boldsymbol{g}(\boldsymbol{x}) = -\frac{\Phi'(m)}{m}\boldsymbol{M}\boldsymbol{x}, \qquad \Phi'(m) \equiv \frac{\mathrm{d}\Phi}{\mathrm{d}m} > 0. \tag{17}$$

Its sky-plane component is

$$\boldsymbol{a}_p(\boldsymbol{x}) = -\frac{\Phi'(m)}{m}\left(\boldsymbol{M}_{pp}\,\boldsymbol{x}_p + z\,\boldsymbol{M}_{pz}\right). \tag{18}$$

Substituting into equation (11) with $\boldsymbol{a}_{p\perp}\cdot\boldsymbol{x}_p = \boldsymbol{a}_p\cdot\boldsymbol{x}_{p\perp}$, the negative prefactor flips the sign:

$$\mathcal{A} = \mathrm{sign}\left[(\boldsymbol{M}_{pp}\boldsymbol{x}_p)\cdot\boldsymbol{x}_{p\perp} + z\,\boldsymbol{M}_{pz}\cdot\boldsymbol{x}_{p\perp}\right]. \tag{19}$$

The first term is non-zero whenever the isopotentials' cross-section in the sky plane is non-circular ($\boldsymbol{M}_{pp} \not\propto \boldsymbol{I}$), so triaxial halos can produce convexities. The second vanishes only when $\hat{\boldsymbol{e}}_z$ is a principal axis of $\boldsymbol{M}$; otherwise it depends on the unobserved LoS position z. Both vanish only for a LoS-axisymmetric potential (Section 3.1).

For a general viewing direction, equation (19) shows that the unknown LoS position z along the projected track prevents us from immediately inferring the shape of an individual halo. Generative stream models can jointly infer that distance track, but conditional on a specific model for the potential (M. A. Fardal et al. 2015; S. Pearson et al. 2022; J. Nibauer & S. Pearson 2026). We can avoid this complication entirely and instead record only whether the projected track contains a convex segment. This motivates considering a population of streams rather than individual cases. Across a population of streams, the frequency of such segments depends on the distribution of halo shapes predicted by the dark matter model. The convexity rate therefore tests a cosmological model without fitting the potential of each host, as we show in Section 5.

4.2. *Convexities in Simulated Streams*

We test the diagnostic on two simulated streams: one particle-spray stream and one N-body stream, each from a different dwarf galaxy progenitor. Both confirm the analytic prediction that large-scale, $> 10kpc$ convexities occur and can be detected in the projected stream tracks. Figure 4 presents these streams, with the top left panel showing the particle-spray stream and the bottom left panel showing the N-body stream. The convex portions are drawn in red, relative to the projected CoM at the cross. We simulate the particle-spray stream in the triaxial NFW potential of Figure 3, with the same initial conditions. To better match the length of a dwarf galaxy stream, we use a 6.5 Gyr integration time with

progenitor mass $M_{\rm prog} = 10^6\,{\rm M}_\odot$. The N-body simulation is drawn from and described in N23, and differs from the particle-spray example in both its potential — here a $q = 1.5$ flattened logarithmic halo — and its initial conditions. The integration time for the N-body simulation is 10.375 Gyr.

For each simulated stream we build a stream track and analyze that track for convexities across viewing directions. To build the stream track we mix methods: doing initial ordering with `potamides` </> (S. Wu et al. 2026), then fitting and outlier removing using `phasecurvefit` </> (N. Starkman 2026), and finally fitting a full 3D ridge line, (J. Nibauer et al. 2022; N. Starkman et al. 2023). From this 3D track we project it onto each viewing direction, and measure the convex arc-length fraction in each 2D projection. The Mollweide plot of Figure 4 shows the convex fraction across viewing directions. The highest convex fractions concentrate near the stream's mean orbital plane, consistent with the analytic prediction that convexity is most visible when the stream is viewed near edge-on in its orbital plane (Section 4.1). Both simulations confirm the predictions of Section 3 and Section 4.1 that the convex arc-length fraction is non-zero in non-spherical potentials and goes to zero in the spherical limit.

5. RESULTS: BUILDING A COSMOLOGY TEST

A single convexity primarily tells us that one halo is not LoS-axisymmetric. Across a galaxy population, the triaxiality of the halos determines the fraction of convexity-containing streams. Since halo triaxiality is a prediction of the dark matter model, so too is the convex stream fraction. In this section we turn the convexity diagnostic into a test of dark matter model predictions.

CDM predicts triaxial halos while self-interactions drive SIDM halos toward sphericity (Section 1). Outside the inner halo the two can therefore have nearly identical radial density profiles but different three-dimensional shapes, even at Milky-Way masses (Y. Fabian Bautista et al. 2025; S. Adhikari et al. 2025). SIDM shapes do gradually approach CDM at large enough radii, as the effects of self-interactions weaken, leaving a radial window whose extent depends on the SIDM cross-section (Section 6.1). The useful radial window is outside the baryon-dominated inner halo but inside the radius beyond which SIDM and CDM shapes converge. Within this window, the models can have similar radial density profiles but different three-dimensional shapes, and convexity measures that shape difference.

Inside the halo scale radius R_s — for NFW halos this is the radius where the logarithmic slope of the density profile is -2, $R_s \approx 15$–$20\,$kpc for galaxies — baryons contribute substantially to the potential and thus also to any measured non-sphericity. Several baryonic scale radii out, the bulge and disk are sub-dominant, and the halo shape reflects the cosmological triaxiality of CDM or SIDM (C. A. Vera-Ciro et al. 2011; J. Baptista et al. 2023). The dwarf galaxy streams that *Euclid*, Rubin, and *Roman* will identify typically lie many scale radii away, far enough that imaging shows the separation. Survey selection therefore puts these streams in the dark-matter-dominated regime.

After we filter out the near-tangent strip where the diagnostic is unreliable (Section 3.2), each stream is just a yes-or-no measurement: does it show a convex segment? In Section 4 we demonstrate that the answer is yes for a small fraction of viewing directions, in potentials motivated by typical CDM galaxy-mass hosts (B. Allgood et al. 2006; C. A. Vera-Ciro et al. 2011; C. Giocoli et al. 2026). The halo-shape distribution determines the rate. Therefore, for a SIDM host of the same mass and radial profile the answer is yes less often than for CDM. Larger self-interaction cross-sections σ/m round the halo out to larger radii, lowering the SIDM convexity rate and widening the difference between the models (Section 6.1; T. Brinckmann et al. 2018; M. S. Fischer et al. 2024). Across a catalog of N streams, the number K of streams with a convexity gives an observed rate $\hat{\lambda}_{\rm obs} \equiv K/N$, which the two dark matter models predict differently.

Section 5.1 motivates the two rates from simplified simulations, establishing their order of magnitude and sensitivity to halo shape. We then use that catalog three ways, each through a different statistic: testing whether it excludes a model outright (Section 5.2, a binomial tail probability), comparing the two models (Section 5.3, a likelihood ratio), and forecasting how many streams a survey needs to reach a given confidence (Section 5.4, a power calculation).

5.1. *The Stream Convexity Rate*

In its most basic form the convexity test needs one number per model cosmology: the fraction of catalog streams in which we detect a convex segment. The catalog contains streams with measured tracks; a stream counts as a detection if, given a CoM, it shows a reliable convex segment (see Section 3.2) that is long enough to recognize, which we quantify below. Let E be that detection event. The population rate of model $\mathcal{M}$ is the probability of E given $\mathcal{M}$, marginalized over the predicted

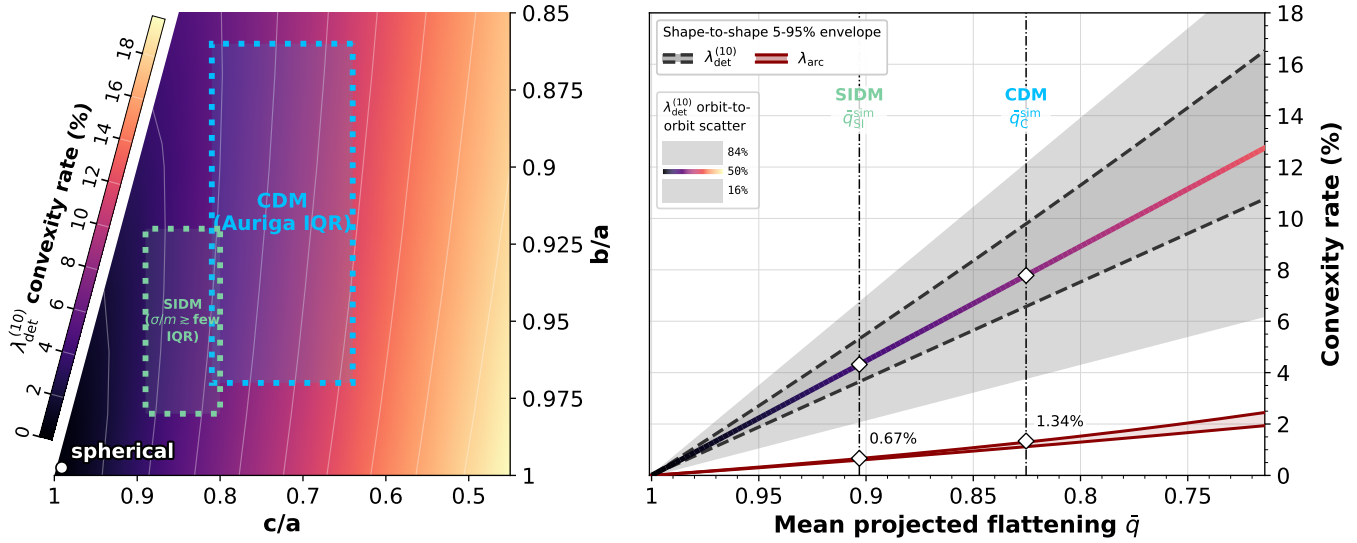


Figure 5. Per-stream convexity rates in a grid of triaxial NFW halo potentials. *Left:* colors show $\lambda_{\rm det}^{(10)}$, the fraction of streams with a convex segment longer than 10 kpc. White contours are lines of constant $\lambda_{\rm det}^{(10)}$ (colorbar). The blank wedge has $b/a < c/a$, excluded by definition. The blue dotted box is the joint interquartile region of the Auriga hydrodynamic CDM halos in $(b/a,\, c/a)$. The green dotted box shows an extrapolated with-baryons SIDM counterpart, obtained by rounding those halo shapes for $\sigma/m \gtrsim$ a few cm^2 g^{-1}. *Right:* the same potentials are plotted against $\bar{q}$, their mean projected minor-to-major density-axis ratio. The colored curve and gray band show the median and orbit-to-orbit spread of $\lambda_{\rm det}^{(10)}$; black dashed curves show how the rate varies among intrinsic halo shapes with the same $\bar{q}$, e.g. the difference between more oblate and more prolate halos. Solid red curves and their shaded envelope show how $\lambda_{\rm arc}$ varies among intrinsic halo shapes with the same $\bar{q}$. The arc-length rate $\lambda_{\rm arc}$ is the fraction of stream length that is convex. It requires convexity over much of a stream, whereas $\lambda_{\rm det}^{(10)}$ requires one resolved segment. A survey's rate should lie between them. Black dot-dashed lines, labeled CDM in blue and SIDM in green, mark the mean $\bar{q}$ of the two populations. At each line, the upper white diamond gives the mean $\lambda_{\rm det}^{(10)}$ and the lower white diamond gives the mean $\lambda_{\rm arc}$. The rounder SIDM population has a lower convexity rate, enabling a statistical test with many streams. Appendix C provides further details.

halo-shape parameters $\boldsymbol{S}$ and the viewing direction $\hat{\boldsymbol{e}}_z$,

$$\lambda_{\mathcal{M}} \equiv P(E \mid \mathcal{M}) \tag{20}$$

$$= \int P(E \mid \boldsymbol{S}, \hat{\boldsymbol{e}}_z)\, P(\boldsymbol{S} \mid \mathcal{M})\, P(\hat{\boldsymbol{e}}_z)\, \mathrm{d}\boldsymbol{S}\, \mathrm{d}\hat{\boldsymbol{e}}_z, \tag{21}$$

where $P(E \mid \boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ is the per-host convexity probability, $P(\boldsymbol{S} \mid \mathcal{M})$ the shape distribution predicted by $\mathcal{M}$, and $P(\hat{\boldsymbol{e}}_z)$ a (generally isotropic) viewing prior, taken independent of $\boldsymbol{S}$. We abbreviate these $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ and $P_{\mathcal{M}}(\boldsymbol{S})$. We label the models $\mathcal{M} \in \{\mathrm{C}, \mathrm{SI}\}\mathrm{DM}$, with convexity rates $\lambda_{\mathcal{M}}$. Because $\lambda_{\mathcal{M}}$ averages over all of $P_{\mathcal{M}}(\boldsymbol{S})$, two models can predict overlapping shape distributions and still predict different convexity rates; we will see this explicitly in Figure 5.

The per-host probability $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ depends on model properties, especially halo shape and assembly history, which influence the distribution of stream progenitors. Cosmological simulations or semi-analytic models can predict this distribution (e.g., A. Dropulic et al. 2025), which we need to test a specific dark matter model. In Figure 5, we show $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ calculated through Monte-Carlo sampling of orbits and viewing angles in a Milky-Way-mass halo. For each potential, we integrate 5000 orbits for 5 Gyr, using apocentres and eccentricities drawn uniformly from 35–100 kpc and $[0, 0.75]$ respectively (E. Sandford et al. 2017), and initial velocity directions drawn uniformly over 0–360° in the local tangent plane. Viewing directions are isotropic, with 7200 external views. We discard projected segments within 10 kpc of the CoM, where galaxy light complicates ridge-line tracing. Each host uses a triaxial NFW density model (`galax.TriaxialNFWPotential`; N. Starkman et al. 2024a), with potential calculated through the Chandrasekhar ellipsoidal formalism (S. Chandrasekhar 1969; J. Binney & S. Tremaine 2008). The density axis ratios are $b/a \equiv r_y/r_x$ and $c/a \equiv r_z/r_x$, with $r_x \geq r_y \geq r_z$. The left panel of Figure 5 shows $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ across a grid of shapes $\boldsymbol{S}$ on the $(b/a,\, c/a)$ plane.

The shape distribution $P_{\mathcal{M}}(\boldsymbol{S})$ is set by the dark matter model, and equation (20) can use any supplied distribution from any model. From cosmological simulations we take the CDM distribution of $(b/a,\, c/a)$ (T. Brinckmann et al. 2018; C. Giocoli et al. 2026), and convert each shape to a viewing-averaged projected flattening

$\bar{q}$. The ratio $\bar{q}$ is the minor-to-major axis ratio of the viewing-direction-averaged projected isodensity ellipse, and depends only on axis ratios (b/a, c/a). These regions describe halos at the radii traced by streams, beyond where baryons dominate the shape (Section 6.1). Self-interaction rates fall with density, so SIDM and CDM halo shapes converge in the outskirts unless σ/m is large (A. H. G. Peter et al. 2013). We therefore use the SIDM region corresponding to $\sigma/m \gtrsim$ a few $\mathrm{cm^2\,g^{-1}}$, for which self-interactions round the halo well beyond the density core; the shape difference from CDM exceeds the density-profile difference out to $r \approx 0.25$–$0.4\,r_{\rm vir}$ (M. S. Fischer et al. 2022, 2024).

At fixed c/a, $\bar{q}$ is not monotonic in b/a. It peaks near $b/a \sim 0.90$–0.94 and then decreases toward the oblate limit, $b/a = 1$. This decrease happens because the projected axis ratio along equatorial sightlines ranges from c/a to c/b. As $b/a \to 1$, this range reduces to c/a. For example, at $c/a = 0.81$, $\bar{q}$ decreases from 0.884 at $b/a = 0.92$ to 0.876 at $b/a = 1$; correspondingly, $\lambda_{\rm det}^{(10)}$ increases from 5.8% to 6.8%. In the oblate limit, $\bar{q} = \int_0^1 \sqrt{(c/a)^2 + [1-(c/a)^2]\,(\cos i)^2}\,\mathrm{d}\cos i$, where i is the inclination angle. This expression agrees with our Monte-Carlo values to better than 10^{-4}. Thus, increasing c/a always increases $\bar{q}$, but increasing b/a decreases $\bar{q}$ beyond the peak. The white contours in the left panel of Figure 5 show this turnover near the oblate limit.

We evaluate the average over the full shape distribution $P_{\mathcal{M}}(\boldsymbol{S})$. For CDM, we use the Auriga halos described in Section 5.1, evaluating from each halo's shape $\bar{q}$ and convexity rate. For SIDM, we use the same halos after relaxation. The two averages therefore compare similarly constructed populations. Dot-dashed lines in the right panel of Figure 5 mark the two population means; Appendix C adds shaded bands for their central 5–95% in $\bar{q}$. As a robustness check, we weight each shape region uniformly instead of using the actual distribution; this gives $\eta = 0.57$ and changes the required N by a factor of 1.42, without changing the order-of-magnitude forecast in Section 5.4.

Any shape distribution, whether predicted theoretically or measured observationally, can enter equation (20) without changing the framework. Public SIDM zoom-in suites now provide shape distributions for several SIDM models (E. O. Nadler et al. 2025). These distributions could replace the relaxation model of Section 5.1 with direct simulation results.

Not every convex segment will be detectable in imaging. Beyond the width and near-tangent conditions of Section 3.2, a segment must be recognizably convex. We investigate that requirement with three estimators, computed on the same orbits and views; survey-specifics such as surface brightness and track fitting we defer to Section 6.4. One measure is the arc-length fraction $\lambda_{\rm arc}$, the convex fraction of the total projected track; it registers a detection only if the convex arcs together cover much of that track, however short or distributed they are. Our second measure, $\lambda_{\rm det}^{(>\ell_{\rm min})}$, subdivides each 5 Gyr orbit into sliding 2 Gyr windows. Each window better matches the lifetime of a dwarf-galaxy stream, and sliding the windows subsamples over the unknown initial orbital phase. Additionally, $\lambda_{\rm det}^{(>\ell_{\rm min})}$ requires a convex segment longer than $\ell_{\rm min}$ for a detection. We will take a conservative $\lambda_{\rm det}^{(10)}$ ($\ell_{\rm min} = 10\,\mathrm{kpc}$) so the convex arc must extend beyond dense baryonic structures like bars or disks. The last measure is just the ideal detection rate $\lambda_{\rm det}^{(0)}$ at the $\ell_{\rm min} \to 0$ limit, assuming convexity is measurable at any length. We expect the real observable rate to lie between $\lambda_{\rm arc}$ and $\lambda_{\rm det}^{(10)}$. For robustness, we adopt the more pessimistic $\lambda_{\rm arc}$ when quoting numbers.

Figure 5 shows $\lambda_{\rm det}^{(10)}$ and $\lambda_{\rm arc}$ across the grid of triaxial NFW density models; Appendix C adds $\lambda_{\rm det}^{(0)}$ and further details. The left panel maps $\lambda_{\rm det}^{(10)}$ over (b/a, c/a). The rate decreases smoothly and nearly monotonically to zero as the density approaches the spherical limit, $(b/a,\, c/a) \to (1, 1)$ (Section 3.1). The two regions in Figure 5 show the joint interquartile regions of the CDM and SIDM halo populations in (b/a, c/a): each rectangle contains half of its population. For CDM, we use the 30 Milky-Way-mass Auriga halos analyzed by J. Prada et al. (2019), pooling their shapes over the radial range traced by streams. Its region covers $b/a = 0.86$–0.97 and $c/a = 0.64$–0.81.

No existing cosmological hydrodynamical suite provides a statistical sample of galaxy-mass (10^{11}–$10^{13}\,\mathrm{M_\odot}$) halos with SIDM cross-sections large enough to round halos at the radii probed by streams, $\sigma/m \gtrsim$ a few $\mathrm{cm^2\,g^{-1}}$. We therefore construct approximate SIDM counterparts to the Auriga hydrodynamic CDM halos by relaxing each axis ratio toward unity, with a scaling as follows. Dark-matter-only simulations at high cross-section set the size of this shift, while lower-cross-section simulations with baryons calibrate the baryon-caused reduction of the halo sphericalization (C. Giocoli et al. 2026). Baryons similarly make CDM and SIDM halo shapes more alike in other simulations (G. Despali et al. 2022; D. Vargya et al. 2022). The halos are relaxed by

$$s_{\rm SI} = s_{\rm C} + f\,(1 - s_{\rm C}), \qquad s \in \{b/a,\ c/a\}, \tag{22}$$

with larger shift for less spherical halos. We choose f so that the median c/a, $\widetilde{c/a} = 0.73$, increases by δ: $f = \delta\,\widetilde{c/a}/(1-\widetilde{c/a}) = 0.441$. We take $\delta = 16.3\%$ from C. Giocoli et al. (2026). Their dark-matter-only halos show a 35.2% offset, while baryons reduce this by a factor of 2.16 over 0.22–0.37 R_{200}, the radial range of r_1 for $\sigma/m \gtrsim$ a few. Thus $\delta = 35.2\%/2.16$, leaving no free parameter. K. T. E. Chua et al. (2021) further suggest that this offset saturates at these cross-sections. This produces the high-σ/m SIDM distribution, whose region covers $b/a = 0.92$–0.98 and $c/a = 0.80$–0.89. This extrapolation is adequate for forecasting how halo sphericalization changes the population convexity rate, but quantitative constraints on σ/m require matched cosmological hydrodynamical CDM and SIDM ensembles in this mass, radial, and cross-section regime. We label the mean projected flattenings of the two populations by $\bar{q}_{\rm C}^{\rm sim}$ and $\bar{q}_{\rm SI}^{\rm sim}$. The forecast uses this population difference, quantified by the rate ratio η.

The right panel of Figure 5 shows the same behavior for $\bar{q}$. The envelopes' vertical spread between potentials at a common $\bar{q}$, is physical. The ratio $\bar{q}$ is only the mean projected flattening, but convexity depends on the full intrinsic shape S. At fixed $\bar{q}$ an oblate halo has a higher convexity rate than a prolate one. The gray percentile bands show the orbit-to-orbit scatter. That scatter is often larger than the separation between neighboring points. Neither a single stream nor a single host can distinguish between models, but the population mean rate can.

Averaging $\lambda_{\rm arc}$ over each population (lower white diamonds) gives $\lambda_{\rm C} \sim 1.3\%$ and $\lambda_{\rm SI} \sim 0.67\%$. With clean data and careful reduction the measured rates might approach $\lambda_{\rm det}^{(10)}$ (upper white diamonds), $\lambda_{\rm C} \sim 7.8\%$ and $\lambda_{\rm SI} \sim 4.3\%$. The ratio of these two rates, $\eta = 0.50$, determines the required catalog size to distinguish between models. Table 1 collects the convexity rates used in this section.

Table 1. Convexity rates used in Section 5.

Symbol	Definition
$\lambda_{\mathcal{M}}$	rate under model $\mathcal{M}$ (equation (20))
$\lambda_{\rm C}$, $\lambda_{\rm SI}$	the $\mathcal{M}$ = CDM and $\mathcal{M}$ = SIDM rates
η	ratio $\lambda_{\rm SI}/\lambda_{\rm C}$
$\lambda_{\rm arc}$	convex arc-length fraction
$\lambda_{\rm det}^{(\ell_{\rm min})}$	windows with a segment longer than $\ell_{\rm min}$
$\lambda_{\rm det}^{(10)}$, $\lambda_{\rm det}^{(0)}$	the $\ell_{\rm min} = 10\,$kpc and $\ell_{\rm min} \to 0$ limits
$\hat{\lambda}_{\rm obs}$	observed rate K/N

5.2. *Excluding a Model*

We now ask whether the observed catalog can exclude either dark matter model. We fix $\lambda_{\rm C}$ and $\lambda_{\rm SI}$ to the simulation-calibrated values from Section 5.1, leaving no free parameters in either model. The catalog likelihood is then binomial, as in other population-proportion measurements based on counts (E. Cameron 2011). Under model $\mathcal{M}$,

$$K \mid \mathcal{M} \sim \mathrm{Bin}(N, \lambda_{\mathcal{M}}). \tag{23}$$

Too many convexities rejects SIDM while too few rejects CDM. Both are binomial tail probabilities,

$$p_{\rm SI}^{+} = P(K' \geq K \mid N, \lambda_{\rm SI}), \quad K' \sim \mathrm{Bin}(N, \lambda_{\rm SI}) \tag{24}$$

$$p_{\rm C}^{-} = P(K' \leq K \mid N, \lambda_{\rm C}), \quad K' \sim \mathrm{Bin}(N, \lambda_{\rm C}), \tag{25}$$

which we report as one-sided Gaussian significances $Z_{{\rm rej},\mathcal{M}} = \Phi^{-1}(1-p)$, with Φ the normal cumulative distribution.

For scaling arguments the normal approximation is convenient,

$$Z_{{\rm rej},\mathcal{M}} \simeq \frac{|\hat{\lambda}_{\rm obs} - \lambda_{\mathcal{M}}|}{\sqrt{\lambda_{\mathcal{M}}(1-\lambda_{\mathcal{M}})/N}}, \tag{26}$$

which shows the significance scaling with N and with the difference between the observed and predicted rates. We do not use the approximation for quoted values. The approximation, called the Wald form, degrades at small λ (L. D. Brown et al. 2002; E. Cameron 2011). Larger N improves but does not fix it. For $\lambda_{\rm SI} = 0.67\%$ it underestimates the required catalog by $\sim 20\%$ at 1σ. We therefore use exact binomial tails for significances and the Wald form for forecasting, where it matches the exact calculation to within 3.1% by 3σ.

Consider a catalog of $N = 3.5 \times 10^3$ streams at the low rate estimate $\lambda_{\rm arc}$ of Section 5.1. If the universe is CDM, the median catalog contains $K = 47$ convexities, and equation (24) rejects SIDM at $p_{\rm SI}^{+} = 9.5 \times 10^{-6}$, or $Z_{\rm rej,SI} = 4.3$. If it is SIDM, the median is $K = 23$, and equation (25) rejects CDM at $p_{\rm C}^{-} = 8.2 \times 10^{-5}$, or $Z_{\rm rej,C} = 3.8$. Equation (26) gives 4.9 and 3.5 for the same two catalogs, overstating the SIDM rejection and understating the CDM one. All four are median expected significances, with half of all catalogs doing worse (G. Cowan et al. 2011). A forecast needs more than the median outcome, and the sample-size calculation of Section 5.4 will set a higher bar, estimating instead $N_\sigma = 2.0$.

5.3. *Favoring a Model*

The same likelihood also compares the two hypotheses directly. When the hypotheses have no free parameters,

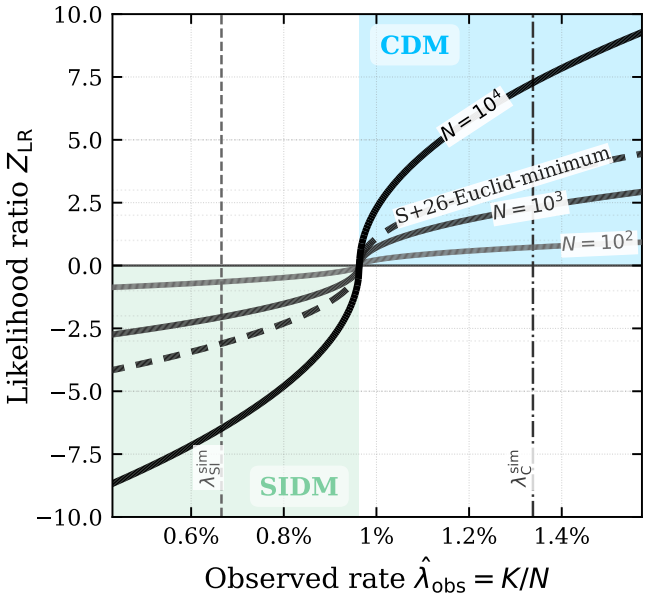


Figure 6. Likelihood-ratio Z_{LR} comparison of CDM and SIDM from the observed rate $\hat{\lambda}_{obs} = K/N$, the K streams containing a convex segment out of a catalog of N, for $N = 10^2, 10^3, 10^4$ (solid). The dashed curve is the $N \approx 2300$ streams projected for *Euclid* by Euclid: Starkman, N. et al. (2026). Curves use the predicted rates $(\lambda_C, \lambda_{SI}) \approx (1.3\%, 0.67\%)$. The statistic Z_{LR} measures relative model preference: positive values favor CDM (blue quadrant), negative values favor SIDM (green quadrant), and larger absolute values mean stronger preference. Shading marks the quadrant in which each model is favored; vertical lines mark the two predicted rates, from Figure 5. With $\sim 10^3$ streams, a rate near either prediction begins to favor its corresponding model; $\sim 10^4$ streams strongly distinguish between the two.

the likelihood ratio is the mathematically optimal test (J. Neyman & E. S. Pearson 1933). We write the log-likelihood ratio with positive values favoring CDM,

$$\Delta \ln \mathcal{L} \equiv \ln \frac{P(K \mid N, \lambda_C)}{P(K \mid N, \lambda_{SI})} = N \left[\hat{\lambda}_{obs} \ln \frac{\lambda_C}{\lambda_{SI}} + (1 - \hat{\lambda}_{obs}) \ln \frac{1 - \lambda_C}{1 - \lambda_{SI}} \right]. \tag{27}$$

At fixed observed rate the log-likelihood ratio scales linearly with N. With a flat prior over {C, SI}DM, the posterior odds in favor of CDM are $\exp(\Delta \ln \mathcal{L})$, the Bayes factor for the pair (R. E. Kass & A. E. Raftery 1995; R. Trotta 2008). Both hypotheses are simple, so the Bayes factor is the likelihood ratio itself, with no evidence integral to evaluate. In Figure 6, we use the signed likelihood-ratio scale (G. Cowan et al. 2011)

$$Z_{LR} = \text{sign}(\Delta \ln \mathcal{L}) \sqrt{2 |\Delta \ln \mathcal{L}|}. \tag{28}$$

The statistic Z_{LR} relabels $\Delta \ln \mathcal{L}$ and is a likelihood ratio, not a significance. The usual factor mapping a likelihood ratio to $N\sigma$ is $\sqrt{2\Delta \ln \mathcal{L}}$, but that conversion needs Wilks' theorem (S. S. Wilks 1938), which does not apply here: CDM and SIDM are not nested models, and both rates are fixed a priori (R. Protassov et al. 2002; S. Algeri et al. 2020). So we use Z_{LR} only to fit a wide range of $\Delta \ln \mathcal{L}$ on one axis, and read the model preference from the actual odds $\exp(\Delta \ln \mathcal{L})$.

Figure 6 shows how to use a catalog to constrain dark matter. For a fixed catalog size, an observed rate near λ_{SI} gives negative Z_{LR} (equation (28)) and favors SIDM, while an observed rate near λ_C gives positive Z_{LR} and favors CDM. Increasing N steepens the curve in Figure 6 because the same observed rate difference carries more binomial information. The horizontal axis is the actual measurement, K/N, so we can read the figure directly once we have a catalog.

5.4. *Distinguishing Dark Matter with Stream Catalogs*

The likelihood of equation (23) fixes the catalog size needed to distinguish the two models at a given significance. We measure that separation through the convexity-rate ratio $\eta \equiv \lambda_{SI}/\lambda_C$, with $\eta \lesssim 1$ since SIDM generally predicts a more spherical halo population.

For a one-sided separation at N_σ, with significance and power set equal ($N_{\sigma,\alpha} = N_{\sigma,\beta} = N_\sigma$), the required sample size in the normal approximation is (e.g., J. L. Fleiss et al. 2003)

$$N \geq N_\sigma^2 \left(\frac{\sqrt{\lambda_C(1-\lambda_C)} + \sqrt{\lambda_{SI}(1-\lambda_{SI})}}{\lambda_C - \lambda_{SI}} \right)^2 \tag{29}$$

$$\gtrsim \frac{N_\sigma^2}{\lambda_C \, (1 - \sqrt{\eta})^2}, \qquad \lambda_C \ll 1 \ (100\%), \tag{30}$$

the second line using $\sqrt{\lambda(1-\lambda)} \simeq \sqrt{\lambda}$. The required N scales inversely with both λ_C and $(1 - \sqrt{\eta})^2$. Taking the low rate estimate of Section 5.1, equation (29) gives $N \sim 850, 3400, 7700, 21\,000$ for $N_\sigma = 1, 2, 3, 5$. The optimistic $\lambda_{det}^{(10)}$ calibration requires a much smaller catalog, $N \sim 180, 740, 1700, 4600$ for the same significances. The suppression ratio is nearly unchanged ($\eta = 0.55$), so a rate 5.8 times larger reduces the required N by close to the same factor, and a 3σ test needs only ~ 1700 streams. Figure 7 plots equation (29) over $(\lambda_C, \eta, N_\sigma)$. The horizontal axis shows λ_C, calibrated in Section 5.1, while η sets the band width. The dependence on η is steep. At the value $\lambda_C \sim 1.3\%$ motivated by Figure 5, moving across the band from $\eta = 0.10$ to $\eta = 0.70$ raises the 3σ requirement from $N \sim 1400$ to $N \sim 25\,000$. The curves in Figure 7 are forecasts, not the significance of an observed catalog. For a measured $K = N\hat{\lambda}_{obs}$, equations (24), (25) and (27) give the model rejection and

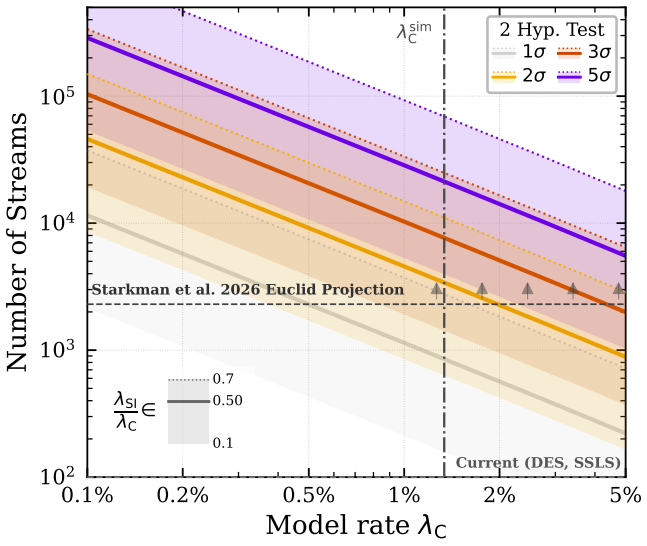


Figure 7. Required stream sample size N to distinguish CDM from SIDM to some σ (equation (29)), as a function of the CDM convexity rate λ_C. Solid lines assume that, as suggested by our simulations, SIDM produces $\eta \equiv \lambda_{SI}/\lambda_C = 0.50$ times the CDM rate of convex streams; shaded bands show the simulated range $\eta \in [0.10, 0.70]$. A larger λ_C or a smaller η (stronger SIDM sphericalization) both reduce the required sample size. A horizontal dashed line with upward arrows marks the lower bound on expected stream-catalog sizes for *Euclid* (Euclid: Starkman, N. et al. 2026); more realistic *Euclid* predictions, and Rubin/*Roman*, expect an order-of-magnitude larger sample. A gray line at $N = 100$ marks approximate current catalog sizes. A vertical dot-dashed line marks the CDM convexity rate, λ_C^{sim}, predicted by our illustrative simulation (Section 5.1). This is a forecast. It neglects observer effects and the cosmological evolution of halo shapes; both modulate the effective λ_C and η. For the simulated convexity rates shown here, large catalogs of streams can distinguish between CDM and SIDM.

model comparison directly, as worked through in Section 5.2.

The vertical band-width in Figure 7 reflects uncertainty in the shape-distribution prediction $P_{\mathcal{M}}(\boldsymbol{S})$, which fixes η. The horizontal position of the dot-dashed line reflects uncertainty in the per-host convexity probability $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$, which fixes λ_C. Reading at the point $(\lambda_C, \eta) \sim (1.3\%, 0.50)$, current Local-Group catalogs ($N \sim 30$–100) remain well below 1σ, and even $N \sim 100$ corresponds to only $\sim 0.34\sigma$. We are currently sample-limited. A 3σ separation between CDM and SIDM is plausible in the *Euclid*/Rubin/*Roman* era, and 5σ needs the largest *Roman*-scale catalogs.

Figure 7 holds λ_C and η fixed, but neither is actually a cosmological constant. The halo-shape distribution $P_{\mathcal{M}}(\boldsymbol{S})$ evolves with redshift, depends on host mass, and for SIDM depends on the cross-section σ/m (C. A. Vera-Ciro et al. 2011; D. Vargya et al. 2022; Y. Fabian Bautista et al. 2025). Hosts are more triaxial at high z and round over cosmic time as adiabatic contraction and elastic scatterings act. A catalog therefore samples many points on this plane rather than one, and its forecast is a survey-weighted average over redshift and host-mass bins. A high-z subsample probes a different point on the (λ_C, η) plane than a $z \sim 0$ subsample. The curves assume that $\|(\boldsymbol{x}_p^{\pm})_\perp\| \gg W$ holds and is well measured, so they are lower bounds for an unselected sample. The survey selection function and the orbital prior both enter $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$, shifting the effective (λ_C, η) away from the simulation calibration and so shifting the curves relative to the survey lines. We discuss this calibration in Section 6.4. Where the forecast falls within that significance band depends on the SIDM microphysics, since the cross-section sets η.

6. DISCUSSION

The convexity criterion of Section 3 works at several scales. For a single stream, from the projected track and CoM a resolved convex segment rules out the whole LoS-axisymmetric centrally-concentrated family of potentials, requiring neither kinematics nor distances (Section 3.1). For a single host galaxy, the fraction of its streams with a convex segment is the per-host probability $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$, evaluated at that host's shape $\boldsymbol{S}$ and LoS $\hat{\boldsymbol{e}}_z$ (Section 5.1). Across a population, averaging $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ over a cosmological shape distribution gives a rate $\lambda_{\mathcal{M}}$ that differs between the CDM and SIDM populations considered here. A stream catalog can therefore test these population-level predictions. In Section 5, we forecast the sample size needed to distinguish them. Here we discuss the connection between this forecast and dark-matter microphysics, including the calibrations required to turn a measured convexity rate into constraints on the self-interaction cross-section (Section 6.1).

6.1. *Convexity Rates Constrain the SIDM Cross-Section*

We now connect the rate forecast of Section 5 to the SIDM self-interaction cross-section. The convexity test directly measures the rate ratio $\eta \equiv \lambda_{SI}/\lambda_C$, not the cross-section. To convert between them, simulations must predict the SIDM halo-shape distribution $P_{SI}(\boldsymbol{S})$ as a function of interaction strength. σ/m parametrizes this strength for velocity-independent models; more generally the velocity-averaged interaction rate $\langle\sigma v\rangle/m$ sets it. This distribution sets λ_{SI} through equation (20), and therefore sets η once the comparator rate λ_C is fixed (for

reviews see S. Tulin & H.-B. Yu 2018; S. Adhikari et al. 2025). Comparing a measured $\hat{\lambda}_{\rm obs}$ with these predicted rates then constrains the range of σ/m allowed by the stream catalog.

Self-interactions redistribute heat and isotropize the velocity tensor on a relaxation time $t_{\rm rel} \sim [\rho \langle \sigma v \rangle / m]^{-1}$ (D. N. Spergel & P. J. Steinhardt 2000). A halo becomes rounder where $t_{\rm rel}$ is shorter than its age and approaches CDM-like triaxiality elsewhere, like at large radii (R. Davé et al. 2001; A. H. G. Peter et al. 2013). In the spherical limit the per-host convexity probability $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$ goes to zero (Section 3). Increasing σ/m therefore rounds a larger part of the halo, suppresses $\lambda_{\rm SI}$, and lowers η. A lower η separates the two model rates more strongly and reduces the number of streams required at fixed significance (Section 5.4).

The mapping from η to σ/m also depends on the radii probed by the streams. We exclude the inner halo, where baryons dominate the mass budget. At large radii, observable streams become increasingly rare. For galaxies, with $M(r < R_{200}) \sim 10^{12}\,{\rm M}_\odot$ and $R_s \approx$ 15–20 kpc, we focus on $30\,{\rm kpc} \lesssim r \lesssim 130\,{\rm kpc}$, the range over which we measure the halo-shape distributions in Section 5.1. The hydrodynamic calibration includes the effect of baryons on halo shape. Hydrodynamic simulations over a wider range of SIDM cross sections would further improve this calibration.

Within this range, the test is informative only where SIDM and CDM predict different shapes. In simulations of galaxies, $\sigma/m \sim 1\,{\rm cm^2\,g^{-1}}$ rounds the inner halo, whereas $\sigma/m \sim 10\,{\rm cm^2\,g^{-1}}$ can extend sphericalization beyond the scale radius (A. H. G. Peter et al. 2013; K. T. E. Chua et al. 2021; M. S. Fischer et al. 2024; Y. Fabian Bautista et al. 2025). Cluster-scale and early-type simulations show the same qualitative behavior (T. Brinckmann et al. 2018; A. Robertson et al. 2019; G. Despali et al. 2022). Streams around disk and early-type galaxies should therefore be constraining for SIDM models where $\sigma/m \gtrsim$ a few ${\rm cm^2\,g^{-1}}$, for which the shape difference to CDM extends beyond the inner galaxy.

Baryons can round a halo at the same radii. In disk galaxies, adiabatic contraction sphericalizes the halo within $r \lesssim 0.1\,R_{200} \approx 20\,{\rm kpc}$ (S. Kazantzidis et al. 2004; O. Sameie et al. 2018; K. T. E. Chua et al. 2022). This contracted region overlaps the region affected by SIDM with $\sigma/m \sim 1\,{\rm cm^2\,g^{-1}}$ (A. H. G. Peter et al. 2013; T. Brinckmann et al. 2018), so contraction and self-interactions can produce similar shape changes in the inner halo. The severity of the degeneracy depends on how far out self-interactions reshape the halo, and simulations do not yet agree. Differences between SIDM and CDM persist farther out in the halo axis ratios than in the spherically averaged density profiles, reaching $r \approx 0.25$–$0.4\,r_{\rm vir}$ (M. S. Fischer et al. 2022). For example, M. S. Fischer et al. (2024) find that a Milky-Way-mass halo with $\sigma/m = 1\,{\rm cm^2\,g^{-1}}$ remains rounder than its CDM counterpart out to $\sim 0.5\,R_{\rm vir}$ in dark-matter-only simulations, beyond which the halo-to-halo scatter exceeds the difference. Stream convexities may therefore remain sensitive to this cross-section. We nevertheless use the more conservative regime in which sphericalization extends beyond the scale radius (C. A. Vera-Ciro et al. 2011), corresponding to $\sigma/m \gtrsim$ a few ${\rm cm^2\,g^{-1}}$ and approximately the $\eta \lesssim 0.5$ part of Figure 7.

Elliptical hosts may extend the test to weaker self-interactions. In the absence of a disk, their CDM halos remain triaxial down to smaller radii (C. A. Vera-Ciro et al. 2011; G. Despali et al. 2022; Y. Fabian Bautista et al. 2025). The usable radial range can therefore extend inward toward R_s, where cross-sections below $1\,{\rm cm^2\,g^{-1}}$ may still alter the halo shape, though halo shapes lose their constraining power as σ/m falls toward $0.1\,{\rm cm^2\,g^{-1}}$ (A. H. G. Peter et al. 2013). Baryons suppress the difference between CDM and SIDM halo shapes at the smaller radii probed by strong lensing and X-ray observations (G. Despali et al. 2022), but stream convexities probe farther out, at 35–100 kpc. Together, disk and elliptical hosts probe $\sigma/m \gtrsim 1\,{\rm cm^2\,g^{-1}}$, and possibly $\sigma/m \sim 0.1$–$1\,{\rm cm^2\,g^{-1}}$ if baryons do not erase the halo-shape difference in elliptical hosts.

For galactic halos, current simulations suggest that the range shown in Figure 7, $\eta \in [0.10, 0.70]$, corresponds roughly to $\sigma/m \sim 1$–$10\,{\rm cm^2\,g^{-1}}$ (D. Vargya et al. 2022; T. Brinckmann et al. 2018; M. S. Fischer et al. 2024; O. Sameie et al. 2021). We therefore interpret our forecasts over this η range as sensitivity to cross-sections of this order.

Stream convexities probe host halos, whose velocity dispersions, $\langle v \rangle \sim 200\,{\rm km\,s^{-1}}$, are well below the cluster-scale dispersions, $\langle v \rangle \sim 1000\,{\rm km\,s^{-1}}$, that set many existing constraints on σ/m. If the SIDM cross-section depends on velocity, stream convexities therefore probe an underexplored interaction regime (S. Tulin & H.-B. Yu 2018; S. Adhikari et al. 2025).

6.2. *Scope and Assumptions*

Throughout this paper, we make no assumptions about the progenitor, but we do assume the following about the host potential and the stream:

Single-component effective potential. A stream may pass near a disk or bar at pericenter, but its convexity at a given location is determined mostly by the local potential rather than by its full orbital history. We therefore restrict the diagnostic to convexity events in the outer halo, where the dark-matter triaxiality signal dominates. For galactic halos, this is roughly $30\,\mathrm{kpc} \lesssim r \lesssim 200\,\mathrm{kpc}$ (Section 6.1).

Time independence and satellite perturbations. The diagnostic assumes that the host potential is approximately static. Massive satellites can violate this assumption by perturbing the stream directly or deforming the host halo, like the Large Magellanic Cloud (LMC) does in the Milky Way (D. Erkal et al. 2019; N. Shipp et al. 2021; N. Garavito-Camargo et al. 2021; S. Lilleengen et al. 2022; R. A. N. Brooks et al. 2025). Such perturbations can create stream convexities in addition to the halo-shape signal. This is consistent with the FIRE-2 forward models of S. Pearson et al. (2026): their long, thin Curlicue stream is recovered without bias in a static potential, whereas the width-varying Massive stream yields a biased halo fit and the S-shaped stream provides no usable constraint outside its progenitor. Therefore, the morphological criteria in Section 3.2 help identify streams for which a static-potential convexity analysis is appropriate. Nearby satellites or visible stream features can flag perturbed systems (A. M. Dillamore et al. 2022; H. R. Foote et al. 2025; A. Bonaca et al. 2020). Nevertheless, isolated streams are the cleanest targets. Future applications should exclude or downweight systems with nearby massive satellites or visible signs of stream perturbations. Section 6.5 discusses their identification in mock extragalactic images.

A known CoM. The test requires the projected CoM of the host because this point defines the radial direction, which in equation (4) determines whether the stream contains convex segments. We take the photometric center as an estimate for the CoM. Any uncertainty or offset between the two can be straightforwardly included by marginalizing $\lambda_{\mathcal{M}}$ over the possible center positions (Section 3).

Stream coherence. We derive the convexity criterion of Section 3 for an orbit but apply it on a stream ridge line. As shown in Section 3.2, the ridge line preserves the local concavity of the underlying orbits when the stream is coherent and its width varies smoothly. These conditions can fail in many places, such as near an intact progenitor, in shell-forming debris, or where the stream fans. Such features are generally recognizable in imaging. A recently dissolved progenitor, however, may leave only a pair of tails. A convexity near the junction of two such tails should similarly be excluded because it comes from the progenitor, not the host potential. Simulated particle-spray and N-body streams support these results (Section 4.2).

6.3. *Related Works*

The closest comparable method is the curvature method of N23. N23 compare the observed curvature to a projected acceleration field from a trial potential. They vary over halo shapes to determine which produces agreement between the stream's track and underlying potential. Our test compares the stream's track to an analytic reference direction ($\bar{\boldsymbol{x}}_p$) supplied by the model class being falsified (here, the LoS-axisymmetric centrally-concentrated family, Section 3.1). Our diagnostic is a limiting case of N23, in which their fitted acceleration becomes analytic and their pointwise likelihood reduces to a binary falsification test. This simplification means we can constrain the maximum flattening, but not the minimum flattening. In trade, our approach needs only one input per stream: whether a convexity exists.

Convex segments are the interesting case. In Figure 5 of N23, stream C contains a convexity and their method correctly rules out a spherical potential in favor of a prolate one. Our approach rules out a spherical potential as well as all LoS-axisymmetric centrally-concentrated potentials simply from the existence of the convex segment. The two tests therefore answer different questions: N23 fit a potential to recover its parameters, while we instead falsify a potential family without fitting any specific potential.

D. Chemaly et al. (2026a) pursue a complementary route through the same stream-shape data. Their hierarchical Bayesian inference returns the full per-host shape posterior, using informative priors and a parametric halo-shape model to extract information from each stream. The method developed here is a much coarser measurement, and it needs significantly larger catalogs for the same constraining power. In trading specificity for volume, our method needs no detailed modeling of individual systems: no galaxy potential model, no careful characterization of each stream beyond whether it has a convexity, and no priors on the progenitor.

A. Bariego-Quintana & F. J. Llanes-Estrada (2024) develop a complementary three-dimensional approach for Milky Way streams. Rather than projected curvature, they use the torsion τ of streams. In a spherical potential orbits are planar and $\tau = 0$. A prolate or cylindrical halo gives a non-zero torsion that grows with the elon-

gation. Applied to Galactic streams, they find torsions consistent with a non-spherical Milky Way halo. The torsion diagnostic requires the full (x, y, z) track and distance information along the stream, so it is applicable to the Milky Way. A. Bariego-Quintana & F. J. Llanes-Estrada acknowledge this directly, noting that at extragalactic distances only the projected curvature is measurable, not the torsion. Our test is a projected counterpart of theirs, and is applicable across large extragalactic samples.

The Sagittarius (Sgr) stream provides two examples of geometric constraints on the Galactic potential. K. V. Johnston et al. (2005) measured orbital-plane precession from the offset between the orbital poles of the leading and trailing tails, constraining the halo flattening. V. Belokurov et al. (2014) measured apsidal precession from the two tails' apocentres, constraining the radial density profile. Both analyses need distances along the stream, whereas our test uses only the projected track.

6.4. *Calibrating the Population Rate*

For the binomial likelihood of equation (23), we use the simulation-calibrated detection rate $\lambda_{\rm det}$: the probability that an observed stream contains at least one detectable convex segment. Figures 5 and 8 show this rate alongside the ideal limit $\lambda_{\rm det}^{(0)}$ and the more conservative arc-length rate $\lambda_{\rm arc}$ (Section 5.1). The halo-shape distributions, the per-host convexity probability, and the selection function are three ingredients of equation (20) that require calibration before equation (30) gives a quantitative bound. Figures 5 and 7 use simple approximations sufficient for the order-of-magnitude forecast here.

The halo-shape distributions. The shape distributions $P_{\rm C}(\boldsymbol{S})$ and $P_{\rm SI}(\boldsymbol{S})$ depend on mass, redshift, and the assumed SIDM cross-section σ/m, and vary with radius even at $z = 0$ (C. A. Vera-Ciro et al. 2011). The relevant mass range is $M_{200} \sim 10^{12}$–$10^{13}\,{\rm M}_\odot$. In this regime, shape differences persist between CDM and SIDM even where the spherically averaged profiles agree (Y. Fabian Bautista et al. 2025; C. Giocoli et al. 2026). In Figure 5, we fix a measured CDM shape distribution and a relaxed SIDM counterpart. This is a reasonable $z \sim 0$ approximation for galaxy-mass halos, but self-interactions weaken with radius, so a precision analysis will require radially resolved shape distributions and stream samples from a new SIDM simulation suite.

The per-host convexity probability $p(\boldsymbol{S}, \hat{\boldsymbol{e}}_z)$. The heat map in Figure 5 Monte Carlo samples representative thin-stream orbits in each triaxial NFW density model, with a Local-Group-inspired orbital prior and isotropic viewing directions. It assumes an NFW host without baryons and does not yet vary the orbital prior across the simulation grid. The gray percentile bands show the resulting orbit-to-orbit scatter.

We can use the observed stream GD-1 (Figure 1) as a qualitative check on this Monte Carlo approach. GD-1 contains a convex segment longer than $\ell_{\rm min} = 10\,{\rm kpc}$ in 3.2% of 10^4 isotropic viewing directions. That fraction lies within the Monte-Carlo predicted range for the different $\lambda_{\mathcal{M}}$ measures, though GD-1 is not drawn from our static host models or orbital prior, so this is a qualitative check only.

The selection function. Both rates, $\lambda_{\rm C}$ and $\lambda_{\rm SI}$, assume all streams are observable, have clear tracks, and that their convex segments lie outside the near-tangent strip ($\|(\boldsymbol{x}_p^{\pm})_\perp\| \gg W$). The selection function for these conditions varies based on the survey, catalog, and stream-detection pipeline (Section 1). The survey-aware statistic measures the rate of observing at least one convexity given that selection function. Computing it requires forward-modeling each candidate catalog (*Gaia*, Rubin/LSST, *Euclid*), which we leave to future work. Our three estimators differ by factors like spatial coverage, track recovery, smoothing, and detection thresholds, modeled through the minimum projected length $\lambda_{\rm det}^{(\ell_{\rm min})}$. Figure 7 uses the fiducial detection rate $\lambda_{\rm det}(10\,{\rm kpc})$ from Figure 5, which applies a minimal selection: a projected length cut and the 10 kpc CoM exclusion. The two rates $\lambda_{\rm arc}$ and $\lambda_{\rm det}^{(0)}$ represent the plausible range for a given survey. A survey approaches $\lambda_{\rm det}^{(0)}$ when short convex segments resolve along the visible track, and $\lambda_{\rm arc}$ when the selection function down-weights the configurations that produce convexity. For full inference, the selection function must be modeled or matched between simulations and observations.

Once we calibrate these ingredients, equation (27) gives the model comparison and equation (30) gives the sample-size requirement. Cosmological evolution of $P_{\mathcal{M}}(\boldsymbol{S})$ implies that a heterogeneous survey requires a weighted integral over redshift- and mass-binned versions of Figure 7, with $\lambda_{\mathcal{M}}$ in equation (29) matched to the observed redshift–mass distribution.

6.5. *Observational Outlook*

We designed the diagnostic for surveys that resolve stream morphology but have no kinematics. *Euclid* (Euclid: Aussel, H. et al. 2025; Euclid: Walmsley, M. et al. 2025) resolves extragalactic streams over a wide

field, Rubin/LSST (Ž. Ivezić et al. 2019) extends Local-Group depth, and *Roman* (D. Spergel et al. 2015; A. Dey et al. 2023) reaches high source densities at small angular scales. None of these surveys delivers proper motions for the extragalactic populations they resolve, so six-dimensional phase space is unavailable for essentially every stream beyond the Local Group, and with it the 3D torsion of A. Bariego-Quintana & F. J. Llanes-Estrada (2024). The curvature diagnostic requires only the projected track and projected CoM.

The stream catalogs needed for this test are already being built. Wide-field imaging searches of nearby galaxies have identified roughly one hundred dwarf stellar streams (D. Martínez-Delgado et al. 2010; D. Martínez-Delgado et al. 2023; D. Martínez-Delgado et al. 2025; J. Miró-Carretero et al. 2024, 2025; E. Sola et al. 2025). Detailed modeling of individual streams has already constrained their host-halo shapes (S. Pearson et al. 2022; M. Walder et al. 2025; J. Nibauer & S. Pearson 2026; S. Wu et al. 2026; D. Chemaly et al. 2026b). Euclid: Starkman, N. et al. (2026) extend these searches to *Euclid* imaging, while Rubin/LSST, *ARRAKIHS*, and *Roman* will find many more streams (Ž. Ivezić et al. 2019; R. Guzman et al. 2022; D. Spergel et al. 2015). Our test applies to globular-cluster and dwarf-galaxy streams, the latter of which are easier to detect in extragalactic imaging.

7. CONCLUSIONS

In this paper we develop a novel framework for using observations of convexities in tracks of stellar streams to constrain the geometry of dark matter halos and distinguish the particle nature of dark matter. We then derive the geometric constraints at the orbit level (Section 3), and carry them to streams via a width-bound geometric argument (Section 3.2). In Section 5 we build a population-level binomial test of CDM against SIDM and its cross-section by comparing the observed convexity rate $\hat{\lambda}_{\rm obs}$ to the predicted CDM and SIDM rates $\lambda_{\rm C}, \lambda_{\rm SI}$. A catalog of convexities in extragalactic streams, like those current large-scale surveys will build, can favor CDM or SIDM by at least 2σ, more likely 3–5σ. Of the stream-based methods, ours assumes the least: no parametric potential, progenitor model, or halo-shape prior.

We summarize the main results below.

- *Geometric falsification.* Detections of stream convexity falsifies any LoS-axisymmetric, centrally-concentrated potential. The result holds at the orbit level (Section 3), for thin streams, and for dwarf-galaxy progenitors via a width-bound geometric argument (Section 3.2).

- *Binomial cosmology test.* Aggregating per-stream measurements across a survey of streams gives a binomial likelihood for CDM vs SIDM halo-shape distributions (Section 5). Samples of $N \sim 3400$–$21\,000$ extragalactic streams suffice for 2σ–5σ at challenging observational rates $(\lambda_{\rm C}, \eta = \lambda_{\rm SI}/\lambda_{\rm C}) \sim (1.3\%, 0.50)$, and possibly as low as ~ 740–4600 with more favorable detection rates $(7.8\%, 0.55)$. These are within reach of catalogs from *Euclid*, *Rubin*, and *Roman*.

- *A cross-section constraint.* The rate ratio $\eta = \lambda_{\rm SI}/\lambda_{\rm C}$ is sensitive to SIDM cross-sections of roughly $\sigma/m \gtrsim 1\,{\rm cm}^2\,{\rm g}^{-1}$. Sensitivity below $\sim 1\,{\rm cm}^2\,{\rm g}^{-1}$ is contingent on CDM and SIDM halos retaining different shapes in elliptical hosts despite the effects of baryons (Section 6.1).

- *Shape-free population test.* The cosmology test needs minimal priors: none about the progenitor and no parametric potential. The data can test any predicted halo-shape distribution. The price of assuming so little is low per-stream signal, so the method needs a large stream catalog.

- *Complementary to higher-signal methods.* Our binary check is parameter-free, and any subsequent potential fit by any other method must be consistent with it.

- *Upcoming survey data.* The diagnostic uses only the projected track and the projected center of mass. It therefore applies directly to the arriving extragalactic stream catalogs from *Euclid*, Rubin, and *Roman*.

Even without detailed modeling of individual streams or their host galaxy potentials, stellar streams in aggregate are a powerful probe of the nature of dark matter.

ACKNOWLEDGMENTS

L.N. received support from the National Science Foundation, via CAREER award AST-2337864. L.N. is also supported by the Sloan Fellowship. · J.N. was supported in part by a National Science Foundation Graduate Research Fellowship, Grant No. DGE 2039656. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation. · S.P. and S.W. were supported by a research grant (VIL53081) from VILLUM FONDEN. This work was also co-funded by the European Union (ERC, BeyondSTREAMS, 101115754) grant. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them. · Support for this work was provided by The Brinson Foundation through a Brinson Prize Fellowship grant to author N.S.

This work made use of Anthropic's Claude (Sonnet and Opus models) (https://www.anthropic.com/claude) as an assistive tool for code development, manuscript editing, suggesting relevant literature, and discussion of conceptual aspects of the analysis. All suggested references were checked against the original sources by the authors, and all scientific content, analysis choices, and conclusions are the authors' own.

Software: `unxt` (N. Starkman et al. 2025), `coordinax` (N. Starkman et al. 2024b), `galax` (N. Starkman et al. 2024a), `phasecurvefit` (N. Starkman 2026), `potamides` (S. Wu et al. 2026), `astropy` (Astropy Collaboration et al. 2022), `equinox` (P. Kidger & C. Garcia 2021), `interpax` (R. Conlin 2023), `jax` (J. Bradbury et al. 2018), `matplotlib` (J. D. Hunter 2007), `numpy` (C. R. Harris et al. 2020), `plum` (W. Bruinsma et al. 2026).

DATA AVAILABILITY

This study does not use observational data; all results are based on the simulations described in the text. Upon publication, we will release the code used to generate every figure and reported value as a reproducible `showyourwork` workflow (R. Luger et al. 2021). In the meantime, the code is available upon reasonable request. The underlying software packages are publicly available (see Software).

AUTHOR CONTRIBUTIONS

(Alphabetical)

L. Necib: Writing – review & editing.

J. Nibauer: Formal analysis, Investigation, Writing – review & editing.

S. Pearson: Writing – review & editing.

N. Starkman: Conceptualization, Methodology, Formal analysis, Software, Investigation, Visualization, Writing – original draft, Writing – review & editing.

S. Wu: Conceptualization, Investigation, Software (supporting), Visualization, Writing – review & editing.

APPENDIX

A. ROBUSTNESS OF THE RIDGE-LINE DIAGNOSTIC

This appendix further develops three aspects of the robustness of the ridge-line diagnostic introduced in Section 3.2. Section A.1 derives the width bound of equation (13) everywhere in the stream, not only at its edges. Section A.2 shows that $\mathcal{A}_{\rm str}$ is not particularly sensitive to how we estimate the ridge line from the imaging. Section A.3 confirms that the orbit-level results of Section 4.1 also apply to the observed stream track.

A.1. *The Width Bound for Every Track*

Equation (13) bounds the ridge-line–orbit tangent mismatch by a width gradient. In Section 3.2 this is done at the stream's edge, but the more general result applies to every track within the stream.

Let a track neighboring the ridge line track have perpendicular offset $\xi(s_p)$, with $|\xi| \leq W$, so that $\boldsymbol{x}_{p_2} = \boldsymbol{x}_p + \xi\,\hat{\boldsymbol{n}}_p$. Differentiating and using equation (2) gives

$$\frac{\mathrm{d}\boldsymbol{x}_{p_2}}{\mathrm{d}s_p} = \left(1 - \kappa_p\,\xi\right)\hat{\boldsymbol{t}}_p + \left(\partial_{s_p}\xi\right)\hat{\boldsymbol{n}}_p, \tag{A1}$$

so the neighboring track tilts from the ridge line by an angle $\delta\theta_{12}$,

$$\tan\delta\theta_{12} = \frac{\partial_{s_p}\xi}{1-\kappa_p\,\xi} \simeq \partial_{s_p}\xi, \tag{A2}$$

where the approximation holds when the stream width is small compared with the local radius of curvature, $\kappa_p W \ll 1$. Setting $\xi = \pm W$ gives the result from Section 3.2, $\tan\delta\theta_{12} = \pm\,\partial_{s_p} W$. Away from major features such as epicyclic folds, the tracks within a stream do not cross, so each track remains at an approximately fixed fraction of the stream width: $\xi = fW$, with $|f| \le 1$ and f varying slowly along the stream. Thus, $\partial_{s_p}\xi \simeq f\,\partial_{s_p} W$, and the tilt of every track relative to the ridge line is bounded by $|\partial_{s_p} W|$, recovering equation (13).

A.2. *Maintaining Curvature Alignment from Physics to Observations*

In this section we show that the difference between the physical mechanisms that produce smooth streams and the observational and numerical means by which we fit a smooth ridge line does not change the diagnostic $\mathcal{A}_{\rm str}$. Considering the spatial aspect of Lagrange-point release, a tidal stream should not be smooth: epicyclic loops and overdensities of wavenumber $k_{\rm epi}$ should structure it about its ridge line (A. H. W. Küpper et al. 2010, 2012). However, the spread in release velocities at the Lagrange-point release mixes the epicyclic phases and damps the loops (J. Bovy 2014) sufficiently that the concavity changes of the loops are not observed. When modeling the stream track from observations, a fit over an along-stream window Δs_p suppresses the loops by $\mathrm{sinc}(k_{\rm epi}\Delta s_p/2)$, so any window with $\Delta s_p \gtrsim 2\pi/k_{\rm epi}$ removes them. These two different mechanisms both suppress small-scale changes in the concavity without impacting any larger-scale changes.

This ridge line modeled from observations will differ from the "true" ridge line, and we want to understand under what conditions their concavities agree. Write the difference between them as a shift $\delta_{\rm sm}\boldsymbol{x}_p$. Let $\boldsymbol{n}_0$ be the ridge-line tangent rotated by $+90°$. Unlike $\hat{\boldsymbol{n}}_p$, which reverses at an inflection point, $\boldsymbol{n}_0$ varies more smoothly along the track. In terms of $\boldsymbol{n}_0$ the alignment equation (4) is

$$\mathcal{A}_{\rm str}(s_p) = -\,\mathrm{sgn}\Big[\big((\boldsymbol{x}_p - \bar{\boldsymbol{x}}_p)\cdot\boldsymbol{n}_0\big)\,\big(\partial^2_{s_p}\boldsymbol{x}_p\cdot\boldsymbol{n}_0\big)\Big], \tag{A3}$$

where the first factor is the signed perpendicular distance from the CoM to the tangent line and the second is the signed curvature. The sign is preserved when $\delta_{\rm sm}\boldsymbol{x}_p$ and $\partial^2_{s_p}\delta_{\rm sm}\boldsymbol{x}_p$ are small compared with the two factors. A shift of order the stream width does not change either, so $\mathcal{A}_{\rm str}$ is the same for both.

The exceptions are when one of the factors vanishes. Section 3.2 already shows these cases are identifiable in imaging, so we exclude them. At an inflection point the curvature vanishes and $\mathcal{A}_{\rm str}$ will change sign; the exact location of the change depends on the ridge line model and does not matter in this work where we only care only that $\mathcal{A}_{\rm str}$ changes. For radial infall the perpendicular distance vanishes (equation (14)), but this configuration does not form streams (D. Hendel & K. V. Johnston 2015). For a line of sight near the orbital plane the projected stream can satisfy these cases, which is why the width conditions of Section 3.2 are important.

A.3. *Streaklines in Triaxial Potentials*

Orbits in triaxial potentials are not uniformly concave, and equation (19) gives $\mathcal{A}_{\rm orb} = \pm 1$ along the track (Section 4.1). We show here that the ridge-line diagnostic $\mathcal{A}_{\rm str}(s_p)$ reproduces that pattern pointwise, so the orbit-level results of Section 4.1 carry over to observed convex segments.

The perpendicular-distance argument applies unchanged, with the orbit-level prediction $\mathcal{A}_{\rm orb}(s_p)$ of equation (19) in place of the constant $+1$. We evaluate it at the position, LoS coordinate, and velocity direction of the star on the ridge line at $(t_0, \tau(s_p))$. Non-axisymmetric forces enter the linearized flow at the same order as axisymmetric ones, so the cold-release scalings $W \sim \delta r$ and $|\partial_{s_p} W| \sim \delta r/r$ carry over. The ridge-line and orbit alignments therefore agree, $\mathcal{A}_{\rm str}(s_p) = \mathcal{A}_{\rm orb}(s_p)$, wherever $\|(\boldsymbol{x}_p^{\pm})_\perp\| \gg \delta r(\tau(s_p))$. Morphological selection (D. Hendel & K. V. Johnston 2015) excludes the near-radial regime as before, so thin streams satisfy this on most of their length. Outside the near-tangent strip the results of Section 4.1 apply to the ridge line, and orbit and ridge statements are interchangeable to $\mathcal{O}(\delta r/r)$.

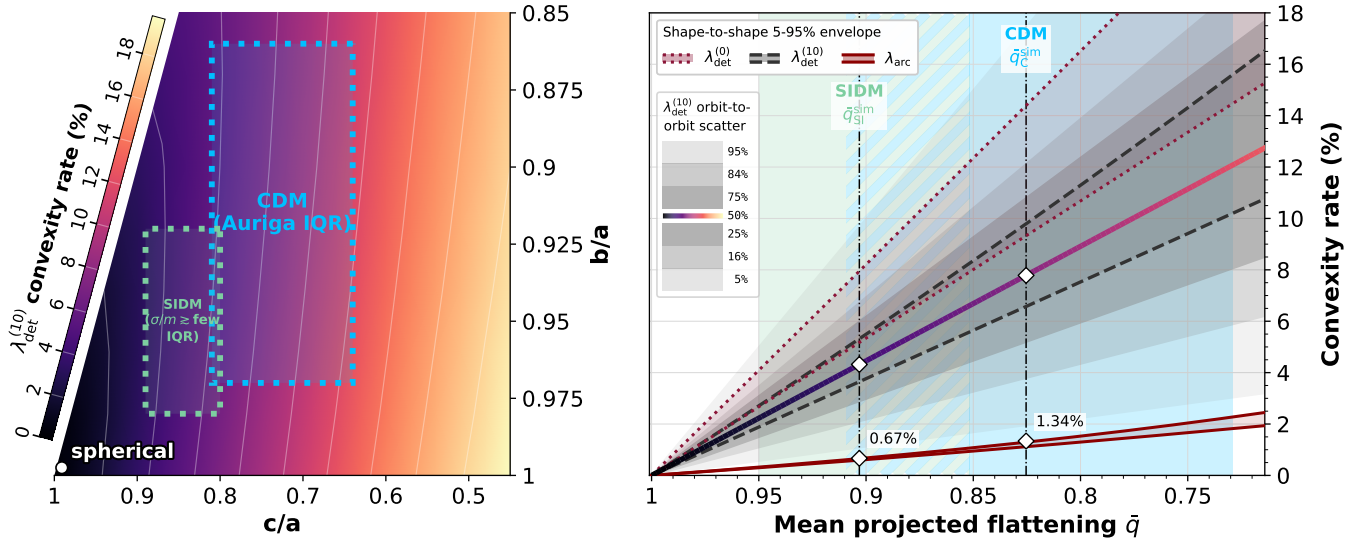


Figure 8. Full version of Figure 5. Burgundy dotted curves add $\lambda_{\rm det}^{(0)}$, the ideal rate if every convex segment is recovered. Nested gray bands show additional orbit-to-orbit ranges. Blue and green vertical bands show the central 5–95% ranges of projected flattening predicted for the CDM and SIDM populations; hatching marks their overlap.

B. TRIAXIALITY IN THE DENSITY VERSUS POTENTIAL

In some sections we work with the potential form $\Phi(m)$ rather than a self-consistent triaxial density $\rho(\boldsymbol{x}) = \rho_0(m)$. For a triaxial density, the potential is typically computed via the Chandrasekhar ellipsoidal integral formalism (S. Chandrasekhar 1969; J. Binney & S. Tremaine 2008). The result has neither the form $\Phi(m)$ nor the closed-form force law of equation (17), so the analytic treatment of Section 4.1 does not apply directly. The gravitational potential is nonetheless always rounder than the density sourcing it (J. Binney & S. Tremaine 2008), so a given density triaxiality produces a weaker triaxiality in the potential. In simulations for Section 5, we instead integrate orbits in the full self-consistent triaxial potential computed from the triaxial NFW density via the Chandrasekhar formalism, with the density axis ratios (b/a, c/a) motivated by cosmological N-body simulations.

C. THE CONVEXITY RATE GRID IN FULL

For readability, Figure 5 omits some of the rate and population-distribution detail. This full version adds $\lambda_{\rm det}^{(0)}$, the ideal rate obtained when every convex segment is recovered; the outer orbit-to-orbit quantile bands; and blue and green vertical bands showing the central 5–95% ranges of projected flattening for the CDM and SIDM populations. Hatching marks where the two projected-flattening ranges overlap.

Because the SIDM population lies close to the spherical limit, its predicted rate depends steeply on $\bar{q}$. Shifting $\bar{q}_{\rm SI}^{\rm sim}$ by 0.0031, well within the variation between different SIDM simulations, changes $\lambda_{\rm SI}$ by $\sim$4%.

REFERENCES

Adhikari, S., Banerjee, A., Boddy, K. K., et al. 2025, Reviews of Modern Physics, 97, 045004, doi: 10.1103/m2vm-59y3

Algeri, S., Aalbers, J., Morå, K. D., & Conrad, J. 2020, Nature Reviews Physics, 2, 245, doi: 10.1038/s42254-020-0169-5

Allgood, B., Flores, R. A., Primack, J. R., et al. 2006, MNRAS, 367, 1781, doi: 10.1111/j.1365-2966.2006.10094.x

Astropy Collaboration, Price-Whelan, A. M., Lim, P. L., et al. 2022, ApJ, 935, 167, doi: 10.3847/1538-4357/ac7c74

Bailin, J., & Steinmetz, M. 2005, ApJ, 627, 647, doi: 10.1086/430397

Baptista, J., Sanderson, R., Huber, D., et al. 2023, ApJ, 958, 44, doi: 10.3847/1538-4357/acea79

Bariego-Quintana, A., & Llanes-Estrada, F. J. 2024, A&A, 687, A46, doi: 10.1051/0004-6361/202347502

Bartelmann, M., & Schneider, P. 2001, Physics Reports, 340, 291, doi: 10.1016/S0370-1573(00)00082-X

Belokurov, V., Koposov, S. E., Evans, N. W., et al. 2014, MNRAS, 437, 116, doi: 10.1093/mnras/stt1862

Bett, P. 2012, MNRAS, 420, 3303, doi: 10.1111/j.1365-2966.2011.20258.x

Binney, J., & Tremaine, S. 2008, Galactic Dynamics, 2nd edn. (Princeton, NJ: Princeton University Press)

Bland-Hawthorn, J., & Gerhard, O. 2016, ARA&A, 54, 529, doi: 10.1146/annurev-astro-081915-023441

Bonaca, A., & Hogg, D. W. 2018, ApJ, 867, 101, doi: 10.3847/1538-4357/aae4da

Bonaca, A., Pearson, S., Price-Whelan, A. M., et al. 2020, ApJ, 889, 70, doi: 10.3847/1538-4357/ab5afe

Bovy, J. 2014, ApJ, 795, 95, doi: 10.1088/0004-637X/795/1/95

Bovy, J., Bahmanyar, A., Fritz, T. K., & Kallivayalil, N. 2016, ApJ, 833, 31, doi: 10.3847/1538-4357/833/1/31

Bradbury, J., Frostig, R., Hawkins, P., et al. 2018, JAX: composable transformations of Python+NumPy programs, 0.10.0 http://github.com/jax-ml/jax

Brinckmann, T., Zavala, J., Rapetti, D., Hansen, S. H., & Vogelsberger, M. 2018, MNRAS, 474, 746, doi: 10.1093/mnras/stx2782

Brooks, R. A. N., Garavito-Camargo, N., Johnston, K. V., et al. 2025, ApJ, 978, 79, doi: 10.3847/1538-4357/ad93a7

Brown, L. D., Cai, T. T., & DasGupta, A. 2002, Annals of Statistics, 30, 160, doi: 10.1214/aos/1015362189

Bruinsma, W., Starkman, N., Vicentini, F., et al. 2026, Plum: Multiple Dispatch in Python, Zenodo, doi: 10.5281/zenodo.5774909

Cameron, E. 2011, PASA, 28, 128, doi: 10.1071/AS10046

Chandrasekhar, S. 1969, Ellipsoidal Figures of Equilibrium (New Haven: Yale University Press)

Chemaly, D., Sola, E., Belokurov, V., et al. 2026a, arXiv e-prints, arXiv:2601.15373, doi: 10.48550/arXiv.2601.15373

Chemaly, D., Sola, E., Koposov, S., et al. 2026b, arXiv e-prints. https://arxiv.org/abs/2607.05510

Chua, K. T. E., Dibert, K., Vogelsberger, M., & Zavala, J. 2021, MNRAS, 500, 1531, doi: 10.1093/mnras/staa3315

Chua, K. T. E., Pillepich, A., Vogelsberger, M., & Hernquist, L. 2019, MNRAS, 484, 476, doi: 10.1093/mnras/sty3531

Chua, K. T. E., Vogelsberger, M., Pillepich, A., & Hernquist, L. 2022, MNRAS, 515, 2681, doi: 10.1093/mnras/stac1897

Conlin, R. 2023, interpax: Interpolation and Function Approximation with JAX, Zenodo, doi: 10.5281/zenodo.10028967

Correnti, M., Bellazzini, M., Ibata, R. A., Ferraro, F. R., & Varghese, A. 2010, ApJ, 721, 329, doi: 10.1088/0004-637X/721/1/329

Cowan, G., Cranmer, K., Gross, E., & Vitells, O. 2011, European Physical Journal C, 71, 1554, doi: 10.1140/epjc/s10052-011-1554-0

Davé, R., Spergel, D. N., Steinhardt, P. J., & Wandelt, B. D. 2001, ApJ, 547, 574, doi: 10.1086/318417

Despali, G., Walls, L. G., Vegetti, S., et al. 2022, MNRAS, 516, 4543, doi: 10.1093/mnras/stac2521

Dey, A., Najita, J., Filion, C., et al. 2023, arXiv e-prints, arXiv:2306.12302, doi: 10.48550/ARXIV.2306.12302

Dillamore, A. M., Belokurov, V., Evans, N. W., & Price-Whelan, A. M. 2022, MNRAS, 516, 1685, doi: 10.1093/mnras/stac2311

Dropulic, A., Shipp, N., Kim, S., et al. 2025, ApJ, 990, 162, doi: 10.3847/1538-4357/adf1a7

Erkal, D., Koposov, S. E., & Belokurov, V. 2017, MNRAS, 470, 60, doi: 10.1093/mnras/stx1208

Erkal, D., Belokurov, V., Laporte, C. F. P., et al. 2019, MNRAS, 487, 2685, doi: 10.1093/mnras/stz1371

Euclid: Aussel, H., Tereno, I., Schirmer, M., et al. 2025, Euclid Quick Data Release (Q1) – Data Release Overview, arXiv, doi: 10.48550/ARXIV.2503.15302

Euclid: Starkman, N., Nibauer, J., Pearson, S., et al. 2026, arXiv e-prints, arXiv:2606.21774, doi: 10.48550/arXiv.2606.21774

Euclid: Walmsley, M., Huertas-Company, M., Quilley, L., et al. 2025, Euclid Quick Data Release (Q1): First Visual Morphology Catalogue, arXiv, doi: 10.48550/ARXIV.2503.15310

Fabian Bautista, Y., Robertson, A., Sagunski, L., Smith-Orlik, A., & Tulin, S. 2025, arXiv e-prints, arXiv:2511.10765, doi: 10.48550/arXiv.2511.10765

Fardal, M. A., Huang, S., & Weinberg, M. D. 2015, MNRAS, 452, 301, doi: 10.1093/mnras/stv1198

Fardal, M. A., Weinberg, M. D., Babul, A., et al. 2013, MNRAS, 434, 2779, doi: 10.1093/mnras/stt1121

Fischer, M. S., Brüggen, M., Schmidt-Hoberg, K., et al. 2022, MNRAS, 516, 1923, doi: 10.1093/mnras/stac2207

Fischer, M. S., Kasselmann, L., Brüggen, M., et al. 2024, MNRAS, 529, 2327, doi: 10.1093/mnras/stae699

Fleiss, J. L., Levin, B., & Paik, M. C. 2003, Statistical Methods for Rates and Proportions, 3rd edn. (Hoboken, NJ: John Wiley & Sons), doi: 10.1002/0471445428

Foote, H. R., Besla, G., Garavito-Camargo, N., et al. 2025, ApJ, 979, 171, doi: 10.3847/1538-4357/ad9b89

Garavito-Camargo, N., Besla, G., Laporte, C. F. P., et al. 2021, ApJ, 919, 109, doi: 10.3847/1538-4357/ac0b44

Giocoli, C., Despali, G., Moscardini, L., et al. 2026, A&A, 706, A340, doi: 10.1051/0004-6361/202558400

Gordon, A. J., Ferguson, A. M. N., & Mann, R. G. 2024, MNRAS, 534, 1459, doi: 10.1093/mnras/stae2169

Grillmair, C. J., & Dionatos, O. 2006, ApJL, 643, L17, doi: 10.1086/505111

Guzman, R., Gómez-Flechoso, M., Martínez-Delgado, D., et al. 2022, in EAS2022, European Astronomical Society Annual Meeting, 1507

Han, J. J., Conroy, C., Johnson, B. D., et al. 2022, AJ, 164, 249, doi: 10.3847/1538-3881/ac97e9

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. 2020, Nature, 585, 357, doi: 10.1038/s41586-020-2649-2

Hendel, D., & Johnston, K. V. 2015, MNRAS, 454, 2472, doi: 10.1093/mnras/stv2035

Holm, J. K., Pearson, S., Nibauer, J., et al. 2026, Nature, 656, 843, doi: 10.1038/s41586-026-10878-w

Hunter, J. D. 2007, Matplotlib: A 2D Graphics Environment, doi: 10.1109/MCSE.2007.55

Ibata, R., Chapman, S., Ferguson, A. M. N., et al. 2004, MNRAS, 351, 117, doi: 10.1111/j.1365-2966.2004.07759.x

Ibata, R., Malhan, K., Tenachi, W., et al. 2024, ApJ, 967, 89, doi: 10.3847/1538-4357/ad382d

Ivezić, Ž., Kahn, S. M., Tyson, J. A., et al. 2019, ApJ, 873, 111, doi: 10.3847/1538-4357/ab042c

Jing, Y. P., & Suto, Y. 2002, ApJ, 574, 538, doi: 10.1086/341065

Johnston, K. V. 1998, ApJ, 495, 297, doi: 10.1086/305273

Johnston, K. V., Law, D. R., & Majewski, S. R. 2005, ApJ, 619, 800, doi: 10.1086/426777

Kaplinghat, M., Keeley, R. E., Linden, T., & Yu, H.-B. 2014, PhRvL, 113, 021302, doi: 10.1103/PhysRevLett.113.021302

Kass, R. E., & Raftery, A. E. 1995, Journal of the American Statistical Association, 90, 773, doi: 10.1080/01621459.1995.10476572

Kazantzidis, S., Abadi, M. G., & Navarro, J. F. 2010, ApJL, 720, L62, doi: 10.1088/2041-8205/720/1/L62

Kazantzidis, S., Kravtsov, A. V., Zentner, A. R., et al. 2004, ApJL, 611, L73, doi: 10.1086/423992

Kidger, P., & Garcia, C. 2021, Differentiable Programming workshop at Neural Information Processing Systems 2021. https://arxiv.org/abs/2111.00254

Koposov, S. E., Rix, H.-W., & Hogg, D. W. 2010, ApJ, 712, 260, doi: 10.1088/0004-637X/712/1/260

Kuhlen, M., Guedes, J., Pillepich, A., Madau, P., & Mayer, L. 2013, ApJ, 765, 10, doi: 10.1088/0004-637X/765/1/10

Küpper, A. H. W., Lane, R. R., & Heggie, D. C. 2012, MNRAS, 420, 2700, doi: 10.1111/j.1365-2966.2011.20242.x

Küpper, A. H. W., Macfarland, T., Heggie, D. C., & Hurley, J. R. 2010, MNRAS, 401, 105, doi: 10.1111/j.1365-2966.2009.15690.x

Law, D. R., & Majewski, S. R. 2010, ApJ, 714, 229, doi: 10.1088/0004-637X/714/1/229

Lilleengen, S., Petersen, M. S., Erkal, D., et al. 2022, MNRAS, 518, 774, doi: 10.1093/mnras/stac3108

Luger, R., Bedell, M., Foreman-Mackey, D., et al. 2021, arXiv e-prints. https://arxiv.org/abs/2110.06271

Malhan, K., Ibata, R. A., Goldman, B., et al. 2018, MNRAS, 478, 3862, doi: 10.1093/mnras/sty1338

Martínez-Delgado, D., Gabany, R. J., Crawford, K., et al. 2010, AJ, 140, 962, doi: 10.1088/0004-6256/140/4/962

Martínez-Delgado, D., Cooper, A. P., Román, J., et al. 2023, A&A, 671, A141, doi: 10.1051/0004-6361/202245011

Martínez-Delgado, D., Stein, M., Sakowska, J. D., et al. 2025, A&A, 701, A182, doi: 10.1051/0004-6361/202554980

Mateu, C. 2023, MNRAS, 520, 5225, doi: 10.1093/mnras/stad321

Miró-Carretero, J., Martínez-Delgado, D., Gómez-Flechoso, M. A., et al. 2024, A&A, 691, A196, doi: 10.1051/0004-6361/202451685

Miró-Carretero, J., Gómez-Flechoso, M. A., Martínez-Delgado, D., et al. 2025, A&A, 700, A176, doi: 10.1051/0004-6361/202452791

Nadler, E. O., Banerjee, A., Adhikari, S., Mao, Y.-Y., & Wechsler, R. H. 2020, ApJ, 896, 112, doi: 10.3847/1538-4357/ab94b0

Nadler, E. O., Kong, D., Yang, D., & Yu, H.-B. 2025, ApJ, 991, 69, doi: 10.3847/1538-4357/adf553

Navarro, J. F., Frenk, C. S., & White, S. D. M. 1996, ApJ, 462, 563, doi: 10.1086/177173

Navarro, J. F., Frenk, C. S., & White, S. D. M. 1997, ApJ, 490, 493, doi: 10.1086/304888

Neyman, J., & Pearson, E. S. 1933, Philosophical Transactions of the Royal Society of London Series A, 231, 289, doi: 10.1098/rsta.1933.0009

Nibauer, J., Belokurov, V., Cranmer, M., Goodman, J., & Ho, S. 2022, ApJ, 940, 22, doi: 10.3847/1538-4357/ac93ee

Nibauer, J., Bonaca, A., & Johnston, K. V. 2023, ApJ, 954, 195, doi: 10.3847/1538-4357/ACE9BC

Nibauer, J., & Pearson, S. 2026, ApJ, 1002, 204, doi: 10.3847/1538-4357/ae5bb0
Pearson, S., Küpper, A. H. W., Johnston, K. V., & Price-Whelan, A. M. 2015, ApJ, 799, 28, doi: 10.1088/0004-637X/799/1/28
Pearson, S., Nibauer, J., Cunningham, E. C., et al. 2026, arXiv e-prints, doi: 10.48550/arXiv.2609.00526
Pearson, S., Price-Whelan, A. M., Hogg, D. W., et al. 2022, ApJ, 941, 19, doi: 10.3847/1538-4357/ac9bfb
Peter, A. H. G., Rocha, M., Bullock, J. S., & Kaplinghat, M. 2013, MNRAS, 430, 105, doi: 10.1093/mnras/sts535
Petit, Q., Ducourant, C., Slezak, E., Sluse, D., & Delchambre, L. 2023, A&A, 669, A132, doi: 10.1051/0004-6361/202244920
Prada, J., Forero-Romero, J. E., Grand, R. J. J., Pakmor, R., & Springel, V. 2019, MNRAS, 490, 4877, doi: 10.1093/mnras/stz2873
Price-Whelan, A. M., & Bonaca, A. 2018, ApJL, 863, L20, doi: 10.3847/2041-8213/aad7b5
Protassov, R., van Dyk, D. A., Connors, A., Kashyap, V. L., & Siemiginowska, A. 2002, ApJ, 571, 545, doi: 10.1086/339856
Robertson, A., Harvey, D., Massey, R., et al. 2019, MNRAS, 488, 3646, doi: 10.1093/mnras/stz1815
Roche, C., McDonald, M., Borrow, J., et al. 2024, The Open Journal of Astrophysics, 7, doi: 10.33232/001c.122309
Sameie, O., Creasey, P., Yu, H.-B., et al. 2018, MNRAS, 479, 359, doi: 10.1093/mnras/sty1516
Sameie, O., Boylan-Kolchin, M., Sanderson, R., et al. 2021, MNRAS, 507, 720, doi: 10.1093/mnras/stab2173
Sanders, J. L., & Binney, J. 2013a, MNRAS, 433, 1813, doi: 10.1093/mnras/stt806
Sanders, J. L., & Binney, J. 2013b, MNRAS, 433, 1826, doi: 10.1093/mnras/stt816
Sandford, E., Küpper, A. H. W., Johnston, K. V., & Diemand, J. 2017, MNRAS, 470, 522, doi: 10.1093/mnras/stx1268
Schaller, M., Robertson, A., Massey, R., Bower, R. G., & Eke, V. R. 2015, MNRAS, 453, L58, doi: 10.1093/mnrasl/slv104
Schrabback, T., Hoekstra, H., Van Waerbeke, L., et al. 2021, A&A, 646, A73, doi: 10.1051/0004-6361/202037670
Shipp, N., Erkal, D., Drlica-Wagner, A., et al. 2021, ApJ, 923, 149, doi: 10.3847/1538-4357/ac2e93
Sola, E., Chemaly, D., Belokurov, V., et al. 2025, MNRAS, 544, 735, doi: 10.1093/mnras/staf1647
Spergel, D., Gehrels, N., Baltay, C., et al. 2015, arXiv e-prints, arXiv:1503.03757, doi: 10.48550/arXiv.1503.03757
Spergel, D. N., & Steinhardt, P. J. 2000, PhRvL, 84, 3760, doi: 10.1103/PhysRevLett.84.3760
Starkman, N. 2026, GalacticDynamics/phasecurvefit: v0.3.1, v0.3.1 Zenodo, doi: 10.5281/zenodo.21345159
Starkman, N., Bovy, J., & Webb, J. J. 2020, MNRAS, 493, 4978, doi: 10.1093/mnras/staa534
Starkman, N., Bovy, J., Webb, J. J., Calvetti, D., & Somersalo, E. 2023, MNRAS, 522, 5022, doi: 10.1093/mnras/stad1166
Starkman, N., Nibauer, J., Bovy, J., et al. 2025, ApJ, 980, 253, doi: 10.3847/1538-4357/ad94f2
Starkman, N., Nibauer, J., & Price-Whelan, A. M. 2024a, galax: Galactic and Gravitational Dynamics in Python, Zenodo, doi: 10.5281/zenodo.11553324
Starkman, N., Nibauer, J., & Price-Whelan, A. M. 2024b, coordinax: Coordinates in JAX, Zenodo, doi: 10.5281/zenodo.10850557
Starkman, N., Price-Whelan, A. M., & Nibauer, J. 2025, unxt: A Python package for unit-aware computing with JAX, doi: 10.21105/joss.07771
Toloba, E., Guhathakurta, P., Romanowsky, A. J., et al. 2016a, ApJ, 824, 35, doi: 10.3847/0004-637X/824/1/35
Toloba, E., Sand, D. J., Spekkens, K., et al. 2016b, ApJL, 816, L5, doi: 10.3847/2041-8205/816/1/L5
Trotta, R. 2008, Contemporary Physics, 49, 71, doi: 10.1080/00107510802066753
Tulin, S., & Yu, H.-B. 2018, PhR, 730, 1, doi: 10.1016/j.physrep.2017.11.004
Valenzuela, L. M., Stoiber, J., & Remus, R.-S. 2026, Deep Imaging Meets Motion: Complementing Stream Photometry Through Planetary Nebula Kinematics, doi: 10.48550/arXiv.2604.14272
van Uitert, E., Hoekstra, H., Schrabback, T., et al. 2012, A&A, 545, A71, doi: 10.1051/0004-6361/201219295
Vargya, D., Sanderson, R., Sameie, O., et al. 2022, MNRAS, 516, 2389, doi: 10.1093/mnras/stac2069
Vasiliev, E., Belokurov, V., & Erkal, D. 2021, MNRAS, 501, 2279, doi: 10.1093/mnras/staa3673
Vera-Ciro, C., & Helmi, A. 2013, ApJL, 773, L4, doi: 10.1088/2041-8205/773/1/L4
Vera-Ciro, C. A., Sales, L. V., Helmi, A., et al. 2011, MNRAS, 416, 1377, doi: 10.1111/j.1365-2966.2011.19134.x
Walder, M., Erkal, D., Collins, M., & Martínez-Delgado, D. 2025, ApJ, 994, 36, doi: 10.3847/1538-4357/adfa1f
Walmsley, M., Allen, C., Aussel, B., et al. 2023, Journal of Open Source Software, 8, 5312, doi: 10.21105/joss.05312
Wilks, S. S. 1938, Annals of Mathematical Statistics, 9, 60, doi: 10.1214/aoms/1177732360

Wu, S., Starkman, N., Pearson, S., et al. 2026, arXiv e-prints. https://arxiv.org/abs/2605.07809